Copernicus Global Land operations
”Vegetation and Energy”
“CGLOPS-1”

Framework Service Contract N° 945120-IPR-2023

ALGORITHM THEORETICAL BASIS DOCUMENT

LAND SURFACE PHENOLOGY

VERSION 1

Issue I1.12

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Dissemination Level		
PU	Public	X
PP	Restricted to other programme participants (including the Commission Services)	
RE	Restricted to a group specified by the consortium (including the Commission Services)	
CO	Confidential, only for members of the consortium (including the Commission Services)	

Document Release Sheet

Book captain:	Lars Eklundh	Sign *Lars Eklundh*	Date 05.05.2025
Approval:	Roselyne Lacaze	Sign *Roselyne Lacaze*	Date 09.05.2025
Endorsement:	Nadine Gobron		
Distribution:	Public		

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Change Record

Date	Page(s)	Description of Change	Release
12.02.2024	All	First version	I1.00
25.03.2024	29, 29	Update Table 2 and parameter in section 3.2.7.1	I1.01
22.04.2024	39, 38 48 - 54	Update comparison with VIIRS VNP22Q2 global phenology products	I1.10
11.07.2024	38	Clarify description of QA in Table 4	I1.11
05.05.2025	28, 29	Add clarification and recommendation in Section 3.2.6 Correct data range in Table 2	I1.12

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Glossary

AD	Applicable document
AMPL	Seasonal amplitude
ASTM	American Society for Testing and Materials
ATBD	Algorithm Theoretical Basis Document
BRDF	Bi-directional Reflectance Distribution Function
CEOS	Committee on Earth Observation Satellites
CGLOPS	Copernicus Global Land Operations
CGLS	Copernicus Global Land Service
CLMS	Copernicus Land Monitoring Service
COG	Cloud-Optimized Geotiff
DL	Double-logistic fitting
DOY	Day of year
DVI	Difference Vegetation Index
ECMWF	European Centre for Medium-Range Weather Forecasts
ERA5	ECMWF Reanalysis v5
ESA	European Space Agency
ESA CCI	ESA Climate Change Initiative
EVI	Enhanced Vegetation Index
EVI2	Two-band Enhanced Vegetation Index
EOS	End of season
EOSD	End of season date
EOSV	Vegetation index value at EOSD
FWHM	Full width at half maximum
GBOV	Ground-Based Observations for Validation
GCC	Green Chromatic Coordinate
GLSP	Global Land Surface Phenology
GPP	Gross Primary Production
HLS	Harmonized Landsat Sentinel-2
HR-VPP	Copernicus High Resolution Vegetation Phenology and Productivity dataset

LAI	Leaf Area Index
LENGTH	Length of season
LPV	Land Product Validation
LSLOPE	Slope of the green-up period
LSP	Land Surface Phenology
MAXD	Day of the maximum of season
MAXV	Maximum value of the season at MAXD
MDVI	Maximum DVI over a time period
MINV	Minimum value of season.
MODIS	Moderate Resolution Imaging Spectroradiometer
MODTRAN	MODerate resolution atmospheric TRANsmission
MR-VPP	EEA Medium Resolution Vegetation Phenology and Productivity dataset
MuSLI	Multi-Source Land Imaging
NASA	National Aeronautics and Space Administration of the USA
NBAR	Nadir BRDF-Adjusted Reflectance
NDVI	Normalized Difference Vegetation Index
NETCDF	Network Common Data Form
NIR	Near-Infrared
OLCI	Ocean and Land Colour Instrument
PPI	Plant Phenology Index
PROBA-V	Project for On-Board Autonomy - Vegetation
PUM	Product User Manual
QA	Quality Assessment
QAR	Quality Assessment Report
QUIL	Quality Index Layer
ReBeLS	Regularised BRDF inversion for Land Surface
RMSD	Root-mean-square deviation
RSLOPE	Slope of the green-down period
S-NPP	Suomi National Polar-orbiting Partnership
SNR	Singal-to-Noise Ratio
SOS	Start of season

SOSD	Start of season date
SOSV	Vegetation index value at SOSD
SP	Spline fitting
SPROD	Seasonal productivity
TIMESAT	a program for analyzing time-series of satellite sensor data
TOC	Top-of-Canopy
TPROD	Total productivity
USA	United States of America
USGS	United States Geological Survey
VIIRS	Visible Infrared Imaging Radiometer Suite
VPP	Vegetation Phenology and Productivity

EXECUTIVE SUMMARY

The Copernicus Global Land Service (CGLS) is earmarked as a component of the Copernicus Land Monitoring Service (CLMS) to operate “a multi-purpose service component” that provides a series of bio-geophysical products on the status and evolution of land surface at global scale. Production and delivery of the parameters take place in a timely manner and are complemented by the constitution of long-term time series.

This Algorithm Theoretical Basis Document (ATBD) describes global land surface phenology (LSP), derived from Sentinel-3 OLCI BRDF-adjusted top-of-canopy (TOC) reflectance data. Land surface phenology is the seasonality of vegetation as observed from satellite instruments. It is described using vegetation phenology and productivity (VPP) parameters, which include e.g., the dates of the seasonal start, maximum and end, seasonal integrals, derivatives, amplitude, and maximum values. A total of 13 VPP parameters, derived for up to two growing seasons per year, make up the CGLOPS LSP product family. The parameters are derived from smooth seasonal trajectories of the plant phenology index (PPI). This vegetation index combines BRDF-adjusted reflectance values in red and near infrared (NIR) bands, and the index is responsive to the seasonal variation in photosynthetically active leaf area index (LAI). This variation is driven by seasonal patterns of solar radiation, temperature and rainfall, and the VPP parameters are thus useful as indicators of climate impacts on vegetation. The product family is generated annually.

1 BACKGROUND

1.1 SCOPE AND OBJECTIVES

The scope of this document is to describe the physical and mathematical basis of the algorithm applied to generate the version 1 of the CGLS Land Surface Phenology product.

This ATBD aims to serve as a reference document for implementing the production chain.

1.2 CONTENT OF THE DOCUMENT

This document is structured as follows:

- Chapter 1 provides an introduction to the document.
- Chapter 2 recalls the user requirements, including the expected performances.
- Chapter 3 describes the retrieval methodology.
- Chapter 4 lists the references cited in this document.
- Chapter 5 contains appendix with additional information.

1.3 RELATED DOCUMENTS

1.3.1 Applicable documents

AD1: Part 2: Technical specifications of Framework Service Contract – Operation of the bio-geophysical variables systematic monitoring of the Global Land Component of the Copernicus Land Service 'CGLOPS' JRC/2023/OP/0273, 19th April 2023.

1.3.2 Input

Document ID	Descriptor
CGLOPS1_ATBD_BRDFCorrection300m-V1	Algorithm Theoretical Basis Document of BRDF correction version 1
CGLOPS1_ATBD_NDVI300m-V2	Algorithm Theoretical Basis Document of the NDVI 300m version 2 product
CGLOPS1_ATBD_LAI300m-V1.1	Algorithm Theoretical Basis Document of Leaf Area Index 300m Version 1.1 product

Documents available at: https://land.copernicus.eu/en/technical-library

1.3.3 Output

Document ID	Descriptor
CGLOPS1_PUM_LSP300m-V1	Product User Manual summarizing all information about the global Land Product Phenology 300m Version 1 product.
CGLOPS1_QAR_LSP300m-V1	Quality Assessment Report of the global Land Product Phenology 300m Version 1 product

Documents available at https://land.copernicus.eu/en/technical-library

1.3.4 External documents

Document ID	Descriptor
GCOS#245	The 2022 GCOS ECVs Requirements https://library.wmo.int/records/item/58111-the-2022-gcos-ecvs-requirements-gcos-245?offset=1

2 USERS REQUIREMENTS

According to the applicable document [AD1], the user's requirements relevant for the Land Surface Phenology (LSP) are described below.

2.1 SPECIFIC TECHNICAL DETAILS AND REQUIREMENTS

PRODUCT SPECIFICATION	
Geometric Properties:	
Spatial resolution	300 m
Geolocation precision	Better than 0.5 pixels
Coordinate position	Centre of the pixel
Geodetic datum	WGS84
Geographic projection	Regular latitude/longitude grid
Geographic coverage	Global
LSP – LAND SURFACE PHENOLOGY	
LSP indicates the timing, date of occurrence, and corresponding value of the vegetation parameter, of key phases of the vegetation development cycle; Phenological metrics related to the start, end of growing season, length of the season and maximum development, seasonal total and partly integrals (as productivity parameters), as well as growth and senescence rates, intra-annual coefficient of variation, etc.	
Time span	Yearly
Timeliness	after one month of last year observation
Uncertainty (2-sigma)	Dates related metrics: Threshold: 10 days, Goal: 5 days
Stability (per decade, i.e. 10 years)	Threshold: 10% Goal: 5%

2.2 FURTHER REQUIREMENTS

2.2.1 Output product composition

Products may contain various information layers and ancillary information – the base reference for product packages are the operational products as on 01.03.2023.

2.2.2 Data Structure

Data coding[1] shall be compatible with the Global Land products as on 01.03.2023 and/or follow the INSPIRE specifications, where applicable.
Ancillary information shall be as currently used and include at least the following:

- The number of measurements per pixel used to generate any synthesis product
- The per-pixel date of the individual measurement or the start-end dates of the period covered
- Quality indicators, with explicit per-pixel identification of the cause of anomalous parameter result.

The product naming and filename conventions that are used in the Copernicus Global Land component production as on 01.03.2023 shall be followed. This may be adapted for complete product collections upon agreement with the contracting authority during the Framework.

2.2.3 Data format

To ensure interoperability with the current Global Land component (operational product data formats and archive data formats) and other Copernicus services, all datasets will be available in NETCDF. Additional format such as Cloud Optimized Geotiff (COG) or ZAR format can be proposed for production or could be requested by the CA during the FWC.

2.2.4 Uncertainties and Validation

Uncertainties indicated in the product specifications above follow the threshold and goal proposed by GCOS#245.

Uncertainties estimates should account for the error propagation uncertainty coming from input data though the retrieval algorithms as during the contract period, ESA plans to imbed uncertainties in the Sentinel-3 ground segment products and, for Sentinel-2, there is an offline tool to determine Evel 1 uncertainties; these can be used for propagation in the production chain.

Validation of the products shall conform to at least the CEOS LPV standards. Wherever appropriate the bio-geophysical variables shall be validated and compared to CEOS CAL/VAL data sets and/or Ground-Based Observations for Validation (GBOV) of Copernicus Global Land component when biophysical parameters are available.

2.2.5 Input data

Copernicus sentinel data are available from https://dataspace.copernicus.eu

Bio-geophysical variable products should be based on common base reflectance data:

[1] Data coding is the provision for the number of bits, range of values, usage of reserved values, content of status map, conversion to physical values formulas.

- Sentinel 3: Derived Top-of-Canopy reflectance may be brokered or produced as it is the case on 01.03.2023 under the CGLOPS contracts;
- Sentinel 2: the Global Land Sentinel-2 Global Mosaic (S2GM)[2] component provides temporal mosaic of surface spectral bands that can be brokered and/or directly used.

Up to and including 2019, products archives of the Global Land component have been based on SPOT VGT, Proba-V, ENVISAT, MODIS, TOPEX/Poseidon, Jason-1, Jason-2, Jason-3, datasets, which are available through the https://dataspace.copernicus.eu.

LST and SWI can be based on geostationary and other satellite data.

Ancillary satellite data that is purchased through Copernicus and put at disposal of the Services is available through the Data Warehouse and will become available on the Copernicus Dataspace Ecosystem.

Ancillary data sets, other than satellite imagery described above, that might be required shall be the responsibility of the contractor.

2.2.6 Product delivery

Products shall be delivered to the Copernicus Land Component dissemination.
The Copernicus Data Space Ecosystem infrastructure will be used to provide access to the final map products.

[2] https://land.copernicus.eu/en/products/global-image-mosaic

3 METHODOLOGY DESCRIPTION

3.1 OVERVIEW

Phenology is the description of recurring biological life cycle events that are linked to variations in weather and climate (Lieth 1974; Schwartz 2013). This connection makes vegetation phenology a crucial indicator in both ecosystem studies and climate change research (Jin et al. 2019; Menzel et al. 2006; Piao et al. 2019; Richardson et al. 2013; Schwartz et al. 2006; Yang et al. 2015; Yu et al. 2010). Plant phenology is also, alongside plant physiological information, a determinant of terrestrial plant productivity (Xia et al. 2015; Zhou et al. 2016), and therefore contributes to the understanding of the terrestrial carbon cycle.

Land Surface Phenology (LSP) is the term for vegetation phenology estimated from remotely sensed data (White and Nemani 2006). It is typically monitored by means of time series of vegetation indices, providing insights into the level of green biomass and photosynthetic activity. LSP estimation depends on the spatial scale of measurement and the plant composition in the studied areas (Helman 2018; Liang and Schwartz 2009). Therefore, LSP estimated from sensors with different spatial resolution will differ.

Several land surface phenology products generated from Earth Observation satellite platforms are currently available and have significantly contributed to research communities and policymaking related to climate change. The objective of this particular product is to sustain its contributions to the community by utilizing the novel Sentinel-3 OLCI data for land surface sensing.

CGLOPS LSP is developed from a vegetation index sensitive to green leaf area, the plant phenology index (PPI). This is a physically based vegetation index, developed from a simplified solution to the radiative transfer equation (Jin and Eklundh 2014). PPI has a linear relationship with green leaf area index (LAI), strong correlation with Gross Primary Production (GPP), and is shown capable of disentangling remotely sensed plant phenology from snow seasonality (Jin et al. 2017; Jin et al. 2019). PPI is reported to be superior to other indices for spring phenology retrieval over the northern latitudes (Karkauskaite et al. 2017) and for GPP estimation in African semi-arid ecosystems (Abdi et al. 2019). Comparison of satellite-derived PPI to ground observations of plant phenology and GPP showed strong similarity of temporal patterns over several Nordic boreal forest sites (Jin et al. 2017). Following an extensive evaluation of PPI against the vegetation indices NDVI and the two-band Enhanced Vegetation Index (EVI2), PPI was chosen as the index to be used in the CLMS HR-VPP project due to its better relationship with ground reference datasets and generally consistent performance (Tian et al. 2021).

To smooth and gap-fill data, LSP parameters are generated from seasonal trajectories developed by time-series smoothing in the TIMESAT software package (Eklundh and Jönsson 2017; Jönsson and Eklundh 2004). CGLOPS LSP seasonal trajectories are based on cubic spline functions, implemented

in TIMESAT, version 4.2. These functions adapt smoothly to seasonal and interannual vegetation index variations. Cubic spline functions have high fidelity and flexibility to represent satellite measurements of plant phenology (Bolton et al. 2020; Cai et al. 2017). The method deals with data gaps in time series of Vegetation Index (VI), and smooths variations due to noise caused by e.g. clouds. The seasonal trajectories from the method are smoothly joined resulting in a continuous function describing the full time period, while capturing the interannual seasonal variations. Bad pixel values due to cloud or snow are weighted down in the function fitting process. An example of a smooth seasonal trajectory fitted to PPI generated from PROBA-V data is shown in Figure 1.

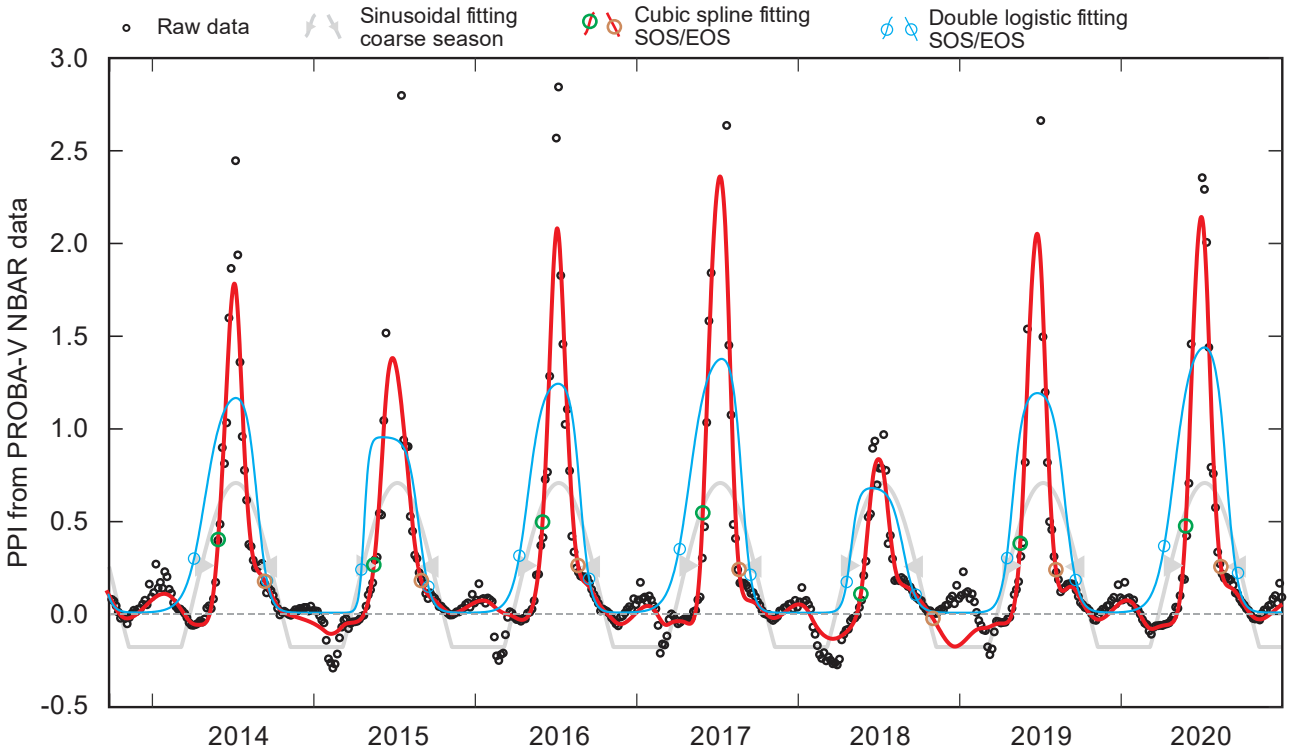


Figure 1. Example of a seasonal trajectory fitted to PPI calculated from PROBA-V Nadir BRDF-Adjusted Reflectances (NBAR) for the testing point EU500P-389 rainfed cropland (60.06626°N, 17.63982°E). Both cubic spline fitting and double logistic fitting were tested in seasonal retrieval, and the cubic spline fitting is adopted here due to its fidelity to raw data. Number of seasons is determined using sinusoidal fitting for coarse season detection.

From the seasonal trajectories (data not stored as output in the CGLOPS product), 13 phenology and productivity parameters for up to two growing seasons per year are extracted (Figure 2).

The computation of start and end of seasons is based on defined threshold points on the smoothed seasonal trajectory.

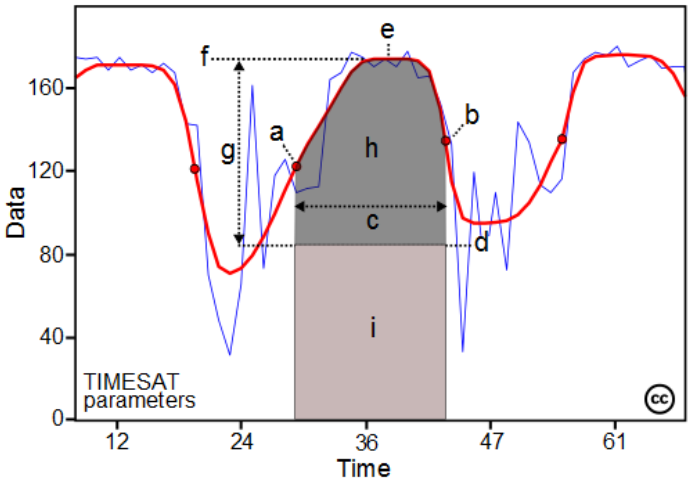


Figure 2. Examples of TIMESAT-derived phenology and productivity parameters: (a) start of season, (b) end of season, (c) length of season, (d) minimum value, (e) time of maximum, (f) maximum value, (g) amplitude, (h) seasonal productivity, (h+i) total productivity. Source: https://web.nateko.lu.se/timesat/timesat.asp.

3.2 THE RETRIEVAL ALGORITHM

3.2.1 Outline

Calculation of LSP products is based on the following steps: (1) Data preparation: generating nadir BRDF-corrected TOC reflectance using ReBeLS NBAR retrieval algorithm and collecting ECMWF ERA5 diffuse fraction of solar radiation. (2) Generation of the plant phenology index (PPI) from Sentinel-3 OLCI TOC BRDF-corrected measurements at 300 m spatial resolution. (3) Processing of TIMESAT version 4.2 to derive smooth seasonal trajectories and phenology and productivity parameters (Figure 3).

For valid land pixels, PPI are computed at five-day time step from BRDF-corrected Sentinel-3 OLCI red and NIR TOC values (NBAR). ERA-5 data of diffuse to total radiation, and maximum difference vegetation index (MDVI) values from the entire NBAR time series from January 2023 to February 2024 are used in the computation (note that during the calibration phase PROBA-V NBAR test data were used from Jan. 2019 to Feb. 2020). The PPI data are fed through to TIMESAT 4.2 for smoothing, computation of phenology and productivity parameters, and derivation of QA flags. The process is further described in section 3.2.7.

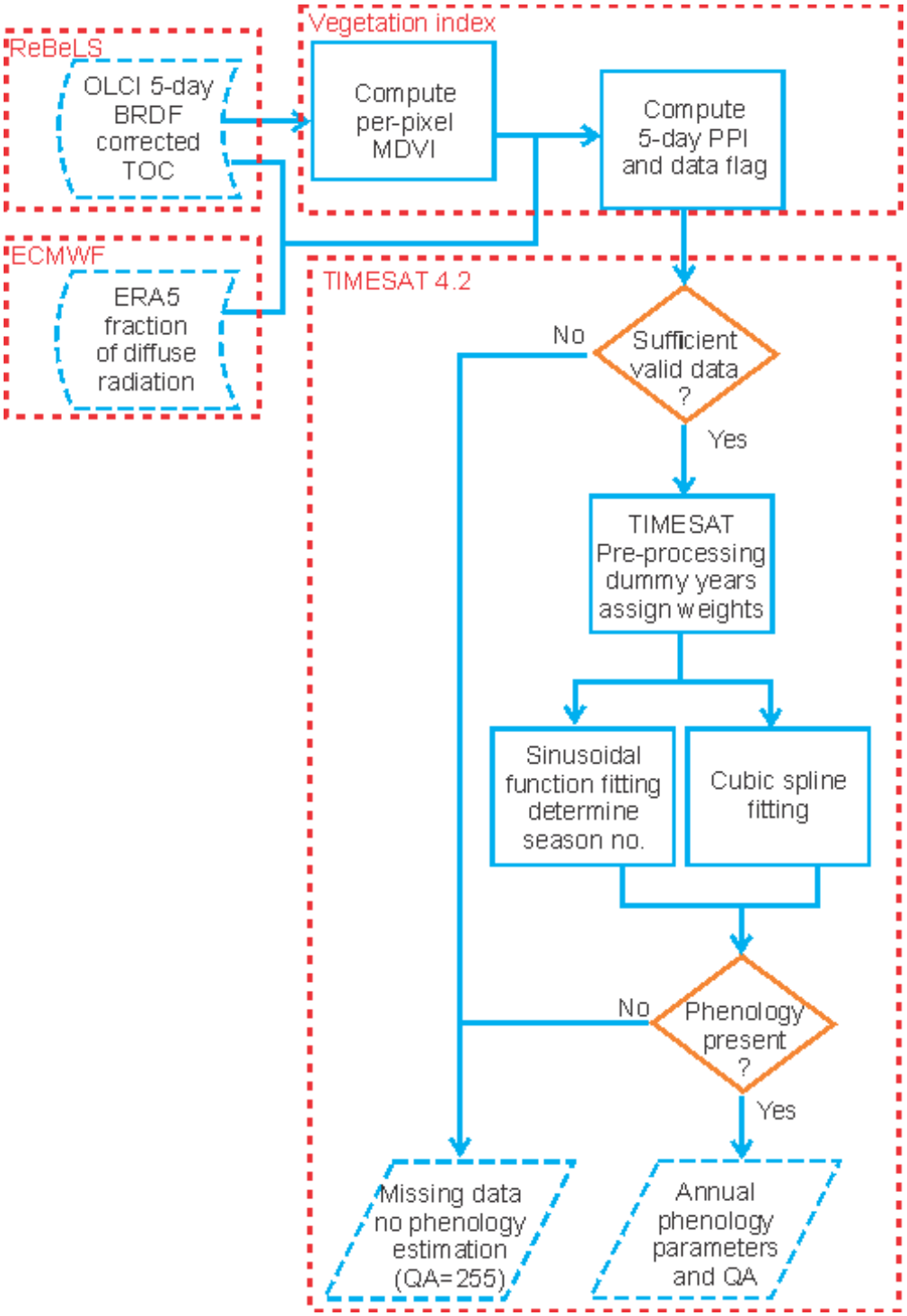


Figure 3. LSP processing overview.

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3.2.2 Basic underlying assumptions

Phenology is determined from smooth seasonal trajectories of vegetation index data, assumed to represent the seasonal development of green canopy foliage. The PPI is developed to represent green leaf area variations with minimal influence of background soil or snow cover. It is assumed that the seasonal PPI trajectories thus primarily represents the seasonal growth curve of the photosynthetically active canopy leaf foliage, including its early-season onset, mid-season peaking and late season growth cessation. It should be noted that no universal criteria for defining the trajectory or points on this line exist. However, it is assumed that by employing a consistent methodology, data can be used for identifying trends and variations in the parameters, thus, enabling conclusions to be drawn regarding the response of the land vegetation to variations in drivers of phenology (weather, climate, management, natural disturbances etc.).

3.2.3 Related and previous applications

This is the first version of CGLOPS LSP. It is modelled on similar principles as developed for the Copernicus Land Monitoring Service *High Resolution Vegetation Phenology and Productivity (HR-VPP)* (https://land.copernicus.eu/pan-european/biophysical-parameters/high-resolution-vegetation-phenology-and-productivity), however at 300 m resolution based on Sentinel-3 OLCI data and with global coverage. Methodological adaptations are made for CGLOPS LSP to account for different input data.

3.2.4 Alternative methodologies currently in use

3.2.4.1 Vegetation indices

The most commonly used vegetation index for estimating vegetation productivity and phenology from the beginning of the satellite era is the normalized difference vegetation index (NDVI) (Tucker 1979). It is e.g., used in the USGS S-NPP eVIIRS 375-m phenology product for the conterminous USA. However, it has been noted that in vegetation types where the seasonal amplitudes are small, and for areas with seasonal snow cover, e.g., in northern coniferous areas, the NDVI is less reliable for phenology retrieval (Delbart et al. 2005; Jin et al. 2017; Jönsson et al. 2010). An alternative index, with better seasonal dynamic in deep vegetation, is the enhanced vegetation index (EVI) (Huete et al. 2002). A two-band version of this index (EVI2) is used in several phenology products, e.g., in the Landsat MuSLI Land Surface Phenology product over the USA (Bolton et al. 2020), and in the 500-m Global Land Surface Phenology (GLSP) product based on VIIRS data (Zhang et al. 2018). To improve mapping of phenology of evergreen coniferous forests and of snow-covered regions, Jin and Eklundh (2014) developed the plant phenology index (PPI). In a comparison against NDVI and EVI2 it was found to be the index that best matched with GPP across Europe (Tian et al. 2021). See Table

1 for a list of currently publicly available LSP products from a breadth of Earth Observation satellite platforms.

Table 1. List of currently publicly accessible remote sensing land surface phenology products.

Product	Region Reso-lution Years	Vegetation indicator	Time series fitting/smoothing methods	Methods for phenology transition	Data source	Data Link	ATBD/User Manual
Global moderate resolution phenology products							
MCD12Q2 (v061)	Global 500 m 2001 – 2022	NBAR-EVI2	Cubic spline fitting (Woltring 1986)	15%, 50%, and 90% of amplitude for green-up and green-down	MODIS	https://e4ftl01.cr.usgs.gov/MOTA/MCD12Q2.061/	https://lpdaac.usgs.gov/documents/1417/MCD12Q2_User_Guide_V61.pdf
VNP22Q2 (v001)	Global 500 m 2013 – 2022	NBAR-EVI2	SG filter, mean/median filter, logistic fitting	Third derivative (extreme curvature change rate)	VIIRS	https://e4ftl01.cr.usgs.gov/VIIRS/VNP22Q2.001/	https://lpdaac.usgs.gov/documents/637/VNP22_User_Guide_V1.pdf
MR-VPP (v01)	Europe 500 m 2000 – 2022	NBAR-PPI	Sinusoidal function fitting + Double-logistic fitting (Zhang et al. 2003)	25% of amplitude for green-up 15% amplitude for green-down	MODIS	https://sdi.eea.europa.eu/catalogue/srv/eng/catalog.search#/metadata/fc62ff38-d77b-452e-9878-acdd14f91b3b	https://land.copernicus.eu/user-corner/technical-library/clms_mrvpp_pum_d1-0.pdf
Regional high resolution phenology products							
MuSLI LSP (v011)	North America 30 m 2016 – 2019	NBAR-EVI2	Cubic spline fitting	15%, 50%, and 90% of amplitude for green-up and green-down	Harmo-nized Landsat 8 – Sentinel-2 (HLS)	https://e4ftl01.cr.usgs.gov/COMMUNITY/MSLSP30NA.011/	https://lpdaac.usgs.gov/documents/1121/MSLSP30NA_User_Guide_V011.pdf
HR-VPP (v01)	Europe 10 m 2017 – 2020	PPI	Cubic spline and double-logistic fitting	25% of amplitude for green-up 15% amplitude for green-down	Sentinel-2A and 2B	https://collections.sentinel-hub.com/tag/phenology/	https://land.copernicus.eu/en/technical-library/product-user-manual-of-vegetation-indices/@@download/file
Others							
HP-LSP (v 1)	Conus US and Alaska 30 m 2019 – 2020	HLS EVI2 + Pheno-Cam GCC	SG filter, mean/median filter, logistic fitting	Third derivative (highest curvature change rate)	Landsat8 Sentinel-2, Pheno-Cam	https://daac.ornl.gov/cgi-bin/dsviewer.pl?ds_id=2248	https://daac.ornl.gov/daacdata/global_vegetation/Landsat8_Sentinel2_Phenocam/comp/Landsat8_Sentinel2_Phenocam.pdf
CLOPS LSP (this product)	Global 300 m 2023	NBAR-PPI	Sinusoidal function fitting (Jönsson and Eklundh 2004) and cubic spline fitting	25% of amplitude for green-up 15% amplitude for green-down	Sentinel-3 and PROBA-V	TBD	TBD

3.2.4.2 Time-series smoothing

Many alternative methods exist for smoothing data and generating seasonal trajectories, and there is no rational basis for selecting a single universal method (White et al. 2009). Some common methods are summarized here. **Double-logistic function fitting** is a very common method in phenology

studies due to its stability to noisy input (Fisher et al. 2006; Jönsson et al. 2018; Zhang et al. 2003). The method is based on non-linear least-squares fitting of functions that describe the generalized seasonal shape of vegetation canopy development. The shape of the functions is determined by parameters controlling the amplitude, width and slope of the curves, and weighted by quality flags of the input data. Double-logistic fitting in TIMESAT 4.1 was applied in the CLMS HR-VPP project due to the irregular nature of the input data from Sentinel-2 (Tian et al. 2021).

Asymmetric Gaussian function fitting (Jönsson and Eklundh 2002) is similar to double logistic function fitting in that it uses least-squares fitting with non-linear parameters, but it has five non-linear parameters, whereas double logistic functions have four, and therefore asymmetric Gaussian functions require more input data to stabilize the fitting. The method is available in TIMESAT 3. **Fourier series** (Menenti et al. 1993; Olsson and Eklundh 1994) and **sinusoidal function fitting** (Jönsson and Eklundh 2004; Roerink et al. 2000) (also called harmonic fitting) uses a series of sinusoidal functions to detect the number of seasons from a noisy time series. The order of the sinusoidal function determines the maximum number of seasons that can be detected. Sinusoidal functions are used here to identify the number and approximate location in time of growing seasons.

When the density and quality of input data are high, several methods that fit more closely to the data are applicable: **Savitzky-Golay filtering** (Chen et al. 2004) smooths time series data by using a least-squares second-order polynomial fit to adjacent values in a window. The window size determines the degree of smoothing. The method is available in TIMESAT. Savitzky-Golay filtering can achieve very good similarity with ground data if it is finely tuned, but there is a risk of producing poor results without tuning (Cai et al. 2017). **Smoothing splines** (Craven and Wahba 1978) can be used to construct continuous daily time series directly. This method can provide smoother curves than Savitzky-Golay, and it produces curves with continuous derivatives everywhere. However, similarly with the problem of selecting window size in Savitzky-Golay, it may be difficult to determine a suitable smoothing parameter that covers all variations across global areas (Cai et al. 2017). **Cubic spline fitting** uses piecewise cubic polynomials to fit time series segments, providing smoother fitting results than lower order spline fitting. Cubic spline fitting can capture the underlying trend of time series and minimize artificial oscillations. The method is available in TIMESAT 4. **Whittaker smoother** is a popular method with similarities to a smoothing spline (Eilers 2003) being flexible to follow rapid variations in data (Atzberger and Eilers 2011).

The nature of input data should determine the selection of time-series processing method. Due to the high frequency of the input data, and following extensive testing, we have selected **cubic spline functions** as the time-series smoothing method to use in CGLOPS LSP (see 3.3.3).

3.2.5 Input data

3.2.5.1 Sentinel-3 Nadir BRDF-adjusted reflectance (NBAR) at red and NIR bands

The spectral bands of Sentinel-3 OLCI are shown in Figure 4 in comparison to other currently widely used satellite sensors in visible and near-infrared spectra for land surface monitoring. There are four narrow red bands (Oa7, Oa8, Oa9, and Oa10) and four narrow NIR bands (Oa16, Oa17, Oa18, and Oa19) potentially useful for vegetation phenology and productivity retrieval. We chose the most suitable red and NIR bands following three common criteria: (1) fine spatial resolution, (2) high signal-to-noise ratio (SNR), and (3) position in high atmosphere transmittance window (less attenuated by atmospheric absorption components). Therefore, the Oa8 (red) and Oa17 (NIR) bands were selected for calculating PPI for the vegetation phenology and productivity estimation in CGLOPS.

Note that this selection of single narrow red and NIR bands for the PPI calculation differs from the CGLOPS NDVI calculation, which averages bands Oa7, Oa8, Oa9 and Oa10 for red reflectance and bands Oa16, Oa17, and Oa18 for NIR reflectance [CGLOPS1_ATBD_NDVI300m-V2]. Solar radiation at the land surface varies with consistently changing atmospheric components (H_2O, CO_2, O_3, and others), wherefore averaging multiple narrow bands to approximate a broadband of a target with varied spectral signatures introduces reflectance uncertainties (Campbell and Norman 1998) that subsequently propagate to the vegetation index. PPI is a physically based vegetation index, and it is sensitive to even slight reflectance changes of dense vegetation during the peak growing season, and is thus sensitive to uncertainties during this time. Therefore, given that the Oa8 and Oa17 bands have high SNR ratios and are located within the high transmittance window of the atmosphere, narrow band averaging is not carried out for the PPI calculation.

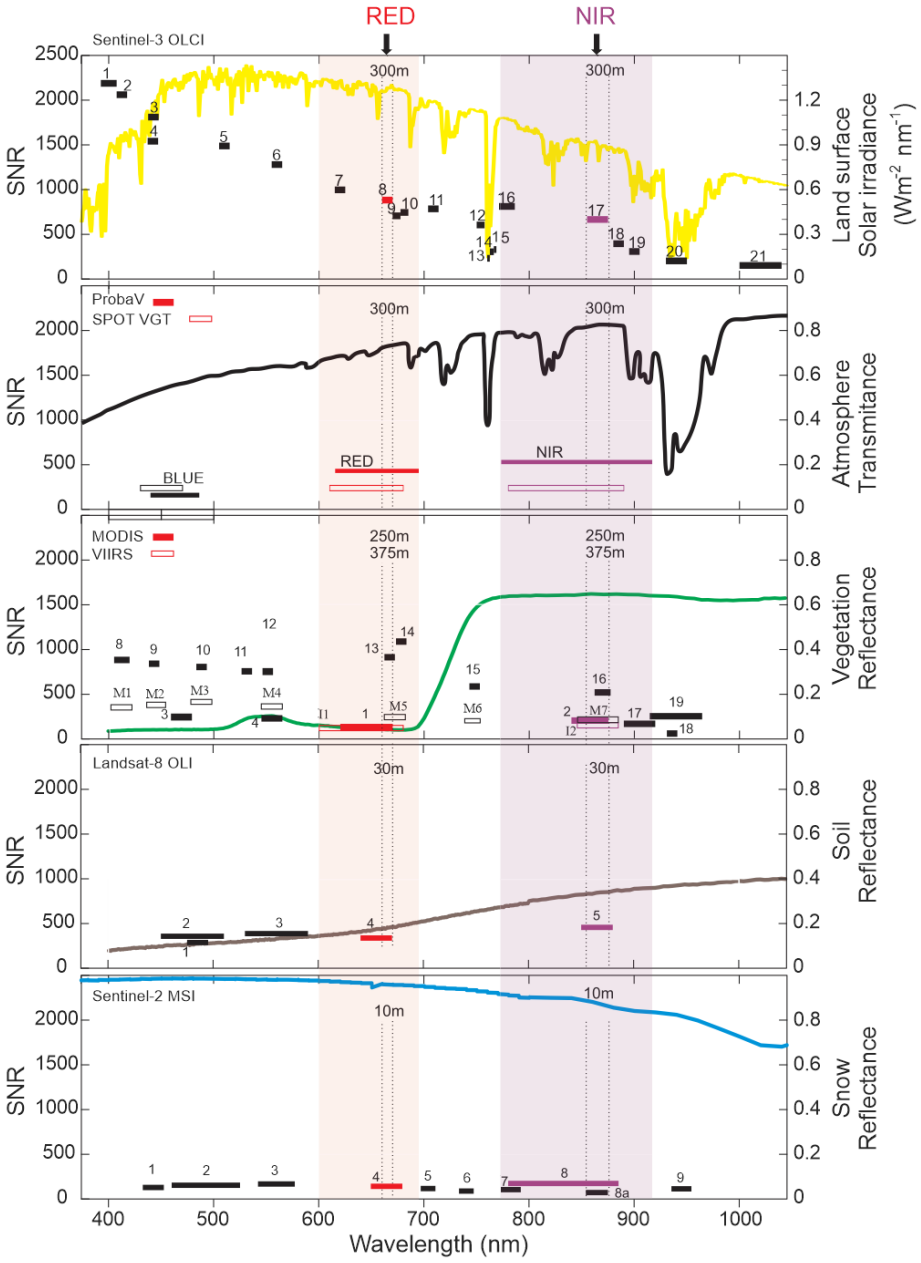


Figure 4. Sentinel-3 OLCI band specification. The horizontal bars denote the bandwidth (FWHM) of the bands, and the y-value is the signal-to-noise ratio (SNR) of the band at reference radiance. Sources: Sentinel-3 band specification: https://sentinel.esa.int/documents/247904/4598066/Sentinel-3-OLCI-Land-Handbook.pdf, PROBA-V band specifications Dierckx et al. (2014), SPOT-5 band specification: https://www.eoportal.org/satellite-missions/spot-5, MODIS and VIIRS Band specifications: https://space.oscar.wmo.int/, Sentinel-2 band specifications: https://sentinels.copernicus.eu/web/sentinel/user-guides/sentinel-2-msi/resolutions/spectral, Landsat 8 band specifications: https://landsat.gsfc.nasa.gov/satellites/landsat-8/spacecraft-instruments/operational-land-imager/oli-requirements/, land surface direct solar irradiance curve: American Society for Testing and Materials (ASTM) G173 (https://www.pveducation.org/pvcdrom/appendices/standard-solar-spectra), atmospheric transmittance: MODTRAN simulation using mid-latitude summer atmosphere model (http://modtran.spectral.com/modtran_home), and spectral signature data of vegetation, soil, and water: Johns Hopkins University Spectral Library (https://speclib.jpl.nasa.gov/documents/jhu_desc).

3.2.5.2 ERA-5 radiation data

The European Centre for Medium-Range Weather Forecasts (ECMWF) Reanalysis v5 (ERA5) single levels provide hourly total sky direct solar radiation at the surface (*fdir*, $J \cdot m^{-2}$) and surface short-wave solar radiation (*ssrd*, $J \cdot m^{-2}$) from 1940 to today at 0.25° × 0.25° resolution (https://cds.climate.copernicus.eu/cdsapp#!/dataset/reanalysis-era5-single-levels). These values at the Sentinel-3 local overpass time 10:00 AM, are downloaded to calculate diffuse fraction of the illumination.

3.2.6 Output product

The vegetation phenology and productivity (VPP) parameters listed in Table 2 are generated.

The physical value range of the Start-Of-Season Date (SOSD) is [-365, 365]. SOSD < 0 indicates that the peak of growing season is in the current year and the start of the season is located in the previous year.

The physical value range of the End-Of-Season Date (EOSD) is [0, 730]. EOSD > 365 indicates that the peak of growing season is in the current year and the end of the season is located in the next year.

It is recommended to use the VPP parameters together with their Quality Assurance (QA) flag (Section 3.2.7.3.7). If only one-year data is used as input, all phenology metrics below 0 and above (365+60) are not reliable (QA=3 poor quality - Table 4). If there are multiple-year data as input, these metrics might be reliable depending on how QA is.

Table 2. Vegetation Phenology and Productivity parameters generated in CGLOPS.

No.	Name	Data type	Scale, offset	Data range	Fill value	Description	Unit
1	SOSD	INT16	1, 0	[-365, 365]	-9999	Day of start-of-season	Day-Of-Year
2	EOSD	INT16	1, 0	[0, 730]	-9999	Day of end-of-season	
3	MAXD	INT16	1, 0	[0, 366]	-9999	Day of maximum-of-season	
4	SOSV	INT16	0.001, 0	[0, 5000]	-9999	Vegetation index value at SOSD	PPI unit $m^2 \times m^{-2}$
5	EOSV	INT16	0.001, 0	[0, 5000]	-9999	Vegetation index value at EOSD	
6	MINV	INT16	0.001, 0	[0, 5000]	-9999	Average vegetation index value of minima on left and right sides of each season	
7	MAXV	INT16	0.001, 0	[0, 5000]	-9999	Vegetation index value at MAXD	
8	AMPL	INT16	0.001, 0	[0, 5000]	-9999	Season amplitude (MAXV – MINV)	
9	LENGTH	INT16	1, 0	[0, 730]	-9999	Length of Season (number of days between start and end)	Day
10	LSLOPE	INT16	0.001, 0	[-1000, 1000]	-9999	Slope of the green-up period	$m^2 \times m^{-2} \times day^{-1}$
11	RSLOPE	INT16	0.001, 0	[-1000, 1000]	-9999	Slope of the green-down period (absolute value of decreasing rate)	
12	TPROD	INT16	1, 0	[0, 2000]	-9999	Total productivity. Growing season integral computed as the sum of all daily values between SOSD and EOSD.	$m^2 \times m^{-2} \times day$
13	SPROD	INT16	1, 0	[0, 2000]	-9999	Seasonal productivity. Growing season integral computed as sum of all daily values minus their base level value.	
Aux	QA	UINT8	-	-	255	Quality flag	-

All output VPP are computed per image pixel and reported to separate output files [CGLOPS1_PUM_LSP-V1].

3.2.7 Detailed algorithm description

3.2.7.1 Calculation of PPI

The calculation of PPI involves three main steps (Figure 5).

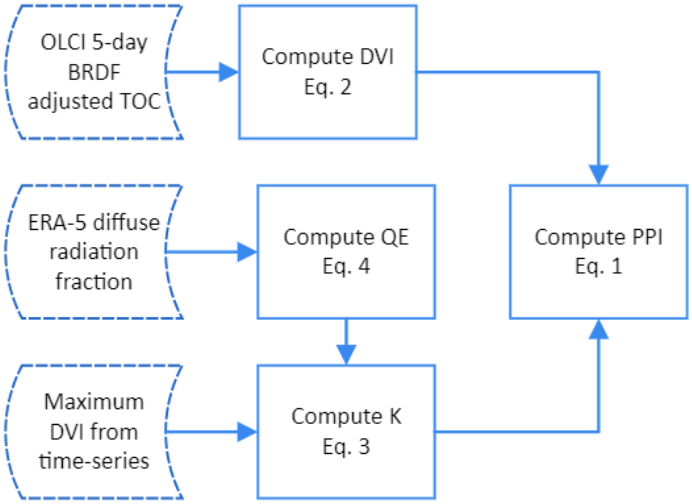


Figure 5. Calculation of PPI.

The formula of PPI (unit: $m^2 \cdot m^{-2}$) is:

$$PPI = -K \times \ln\left(\frac{MDVI - DVI}{MDVI - DVI_s}\right), \qquad Eq.1$$

where DVI is the difference vegetation index. DVI is a dimensionless index and is obtained by subtracting the red TOC reflectance from the near-infrared (NIR) TOC reflectance,

$$DVI = R_{NIR} - R_{RED}\,. \qquad Eq.\ 2$$

$MDVI$ is the temporal potential maximum DVI, which represents infinite leaf layers for a pixel with leaf traits that suit the environmental conditions of the location. $MDVI$ can be estimated from long-term satellite estimation, for example, MODIS Nadir BRDF-Adjusted Reflectance (NBAR) data of over 20 years, which are used in EEA Medium Resolution Vegetation Phenology and Productivity dataset (MR-VPP). Considering the short period of Sentinel-3 observation, we may estimate $MDVI$ from about 7 years of PROBA-V data. The MDVI estimation from the maximum value of a moderately long (5+ years) time series of DVI works well in densely vegetated areas due to the asymptotic nature of vegetation spectral reflectance (Tucker 1977). However, in sparsely vegetated areas, the asymptotic MDVI level cannot be reached. To address this issue, a lower boundary value of 0.18 is assigned as the MDVI value for pixels below this threshold. This value represents the mode value of DVI across 297 PROBA-V tiles globally for 2018-2019, providing an enhanced spatial consistency of LSP productivity. See Figure 12 in Appendix 5.1 for an example of the PPI growing season sum over the Sahelian and Arabian Deserts. DVI_s is introduced in Eq.1 to minimize influences of soil background on vegetation signals, and meanwhile to capture sparse vegetation areas such as drylands. The steps for estimating $MDVI$ and DVI_s are $MDVI = \max(DVI)$, from PROBA-V data

from 2014 to 2021. $MDVI = 0.18, for\ MDVI < 0.18$. $DVI_s = \min(0.09, MDVI_{scene}/4)$, and $MDVI_{scene} = \max(MDVI)$ over a scene. The K factor in PPI is a gain factor given by

$$K = \frac{1}{4Q_E} \cdot \frac{1 + MDVI}{1 - MDVI}, \qquad Eq.3$$

where Q_E is the canopy leaf light extinction efficiency, the ratio of leaf cross-section area on the light beam to leaf geometric cross-section area, unit: $m^2 \cdot m^{-2}$ (Hapke 1993). Q_E is related to leaf inclination angle, solar angle, and the diffuse fraction of solar radiation:

$$Q_E = d_c + (1 - d_c) \cdot \frac{G}{\cos(\theta_i)}, \qquad Eq.4$$

where G is a geometric function of leaf angular distribution, set to 0.5, and d_c is an instantaneous diffuse fraction of solar radiation of the day and local overpass time of PROBA-V and Sentinel-3, i.e. 10:00 AM. The total sky direct solar radiation at the surface (*fdir*, $J \cdot m^{-2}$) and surface short-wave solar radiation downwards (*ssrd*, $J \cdot m^{-2}$) at 10:00 AM are obtained from the European Centre for Medium-Range Weather Forecasts (ECMWF) Reanalysis v5 (ERA5) single levels data.

$$d_c = 1 - \frac{fdir}{ssrd}. \qquad Eq.5$$

The d_c calculated from ERA5 at 0.25° x 0.25° resolution is downscaled to Sentinel-3 resolution (300 m) using bilinear interpolation. For the pixels without ERA5 data available, a backup algorithm for diffuse fraction at solar zenith angle θ_i for clear sky of standard atmosphere conditions is used:

$$d_c = 0.0336 + \frac{0.0477}{\cos(\theta_i)}. \qquad Eq.6$$

The range of PPI is restricted between -1 and 5 to reduce the noise sensitivity in high PPI values. Values below -1 are set to -1 and values above 5 are set to 5. We keep the negative PPI values over non-vegetated areas (water, snow, bare ground, etc.) for later potential use in exploring land surface properties other than vegetation.

3.2.7.2 Calculation of PPI Data flag (DFlag) values for TIMESAT processing

The PPI data quality is estimated from PROBA-V (Sentinel-3) NBAR red and NIR bands 8-bit QUIL quality dataset [CGLOPS1_ATBD_BRDFCorrection300m-V1], as a weighted quality of QUIL values of red and NIR bands, with higher weight (0.9) assigned to NIR, and lower weight (0.1) to red. This is because the NIR band contains the majority of information of the vegetation canopy and the red band only partly contains vegetation information, due to the fact that NIR light easily can penetrate through a dense canopy and be reflected back to satellite sensors, while red light hardly

penetrates through a dense canopy and is only reflected back with information from the canopy surface. PPI data quality flag values and weights for TIMESAT processing are listed in Table 3.

Table 3. PPI data quality flag values and weights for TIMESAT

DFlag	PPI quality	Weight for TIMESAT
0 – 5	Best	1
6 – 9	Good	0.6
10 – 31	Fair	0.1
≥ 32	Poor	0

The output PPI dataset information is shown in Table 5 of Appendix 5.2. Note that PPI data are not planned for further storage or public access.

3.2.7.3 Extraction of phenology and productivity parameters using TIMESAT

An overview of TIMESAT version 4.2 processing is shown in Figure 6.

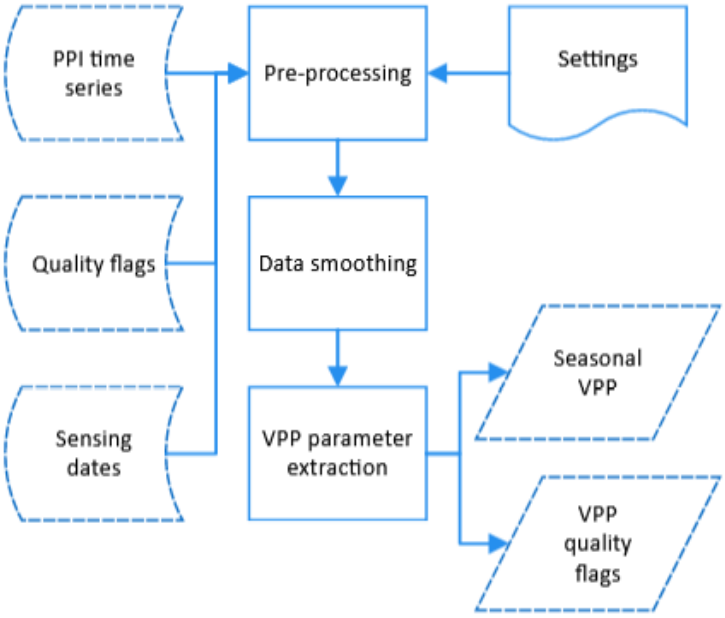


Figure 6. TIMESAT processing overview.

3.2.7.3.1 Pre-processing

TIMESAT receives PPI time-series data that may include some noise and temporal gaps. The pre-processing is the first processing step for the input data, containing several parts: determining the *base level*, setting *data weights*, adding *control points to data gaps*, and determining the *coarse growing seasons*. The pre-processing has four purposes: (1) to transfer the input data (PPI and DFlag data) to pixel-oriented time series for the later processing; (2) to fill large data gaps exceeding 91 continuous days with a predefined base level; (3) to preliminarily determine whether a time series needs to be excluded from processing; and (4) to make a preliminary analysis of seasonality (number of potential seasons) and to establish the rough timing of each coarse season.

The 91-day gap detection and filling mainly address the high-latitude winter seasons that are masked out in the BRDF processing [CGLOPS1_ATBD_NDVI300m-V2]. The base level is determined as the 5th percentile of the good quality PPI observations (weight > 0) histogram for the complete time period. The weights of PPI values in a time series are set based on DFlag data. Further, weights of PPI observations that are lower than the base level are set to 0. The base level is used for filling long term data gaps and assisting the cubic spline fitting. TIMESAT will detect whether there are enough good-quality data on the time series, see Section 3.2.7.3.6 for details. This step is to ensure the operation of fitting the cubic spline functions.

Pixel-wise sensing dates of Sentinel-3 reflectance values are converted to a sequential time vector, starting from the first day of the first year in the input period, e.g., 1 corresponding to 2023-01-01 if the input starts from 2023. To extract the phenological information of the first and last seasons from the time series, the past practice in TIMESAT version 3 was to extend the length of the time series by duplicating the data of the first and last years (Jönsson and Eklundh, 2004; Eklundh and Jönsson, 2017). But, when the input data for the first year starts e.g. in October, such an operation would generate data gaps from January to September. Therefore, before copying the first year, TIMESAT 4.2 fills in the data of the first year by using data from the following year, and then performs the copy operation. The corresponding operation is applied for the last year.

3.2.7.3.2 Daily PPI construction

The PPI time series is fitted with a cubic smoothing spline $S_p(\mathrm{t})$ (Craven and Wahba 1978), which minimizes the value of a criterion function C_p

$$C_p = \sum_{i=1}^{n} \left\{ w_i \left[y_i - S_p(t_i) \right]^2 + p \int_{-\infty}^{+\infty} \left| S_p''(t) \right|^2 dt \right\}, \qquad Eq.7$$

where $t_i, i = 1, \ldots, n$ is time vector and y_i are corresponding PPI values. Each point in the time series is associated with an input weight w_i. The smoothing parameter p controls the shape of the spline,

varying from an exactly interpolating spline ($p = 0$) to a straight line ($p \to \infty$). The p value is set to 1,000, which can reduce the impact of noise while preserving sufficient local variation.

3.2.7.3.3 Extraction of phenology and productivity parameters

An initial pixel-wise estimate of the growing seasons' start and stop dates ("coarse seasons") is determined by applying a sinusoidal function fitting

$$f(t) = c_1 + c_2 \sin(\omega t) + c_3 \cos(\omega t) + c_4 \sin(2\omega t) + c_5 \cos(2\omega t), \qquad Eq. 8$$

where $\omega = 2\pi/n$, and n is the number of points, $f(t)$ is the corresponding PPI value at time t. The coarse seasons are further used for defining potential seasons.

Extraction of VPP parameters from the daily PPI time series shares the same principles as the MCD12Q2 and the HLS phenology products (Bolton et al. 2020; Gray et al. 2019). The VPP processing receives smooth function fits to PPI at daily time step, extracting VPP parameters, checking VPP parameters, and assigning seasons to years (Figure 7). Thirteen VPP parameters and seasonal VPP parameters qualities are generated from the processing, for delivery to users (Figure 2; Table 2).

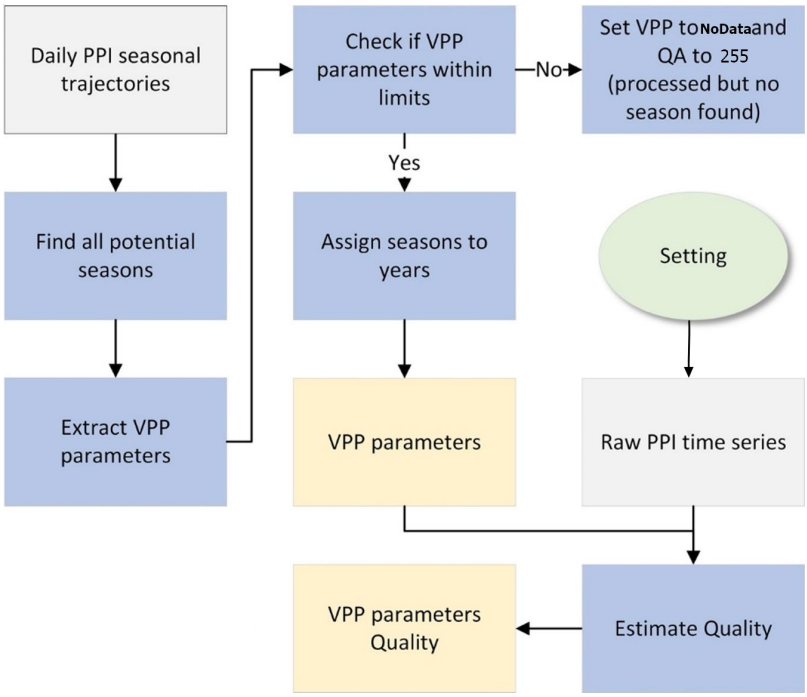


Figure 7. TIMESAT module for extracting phenological parameters.

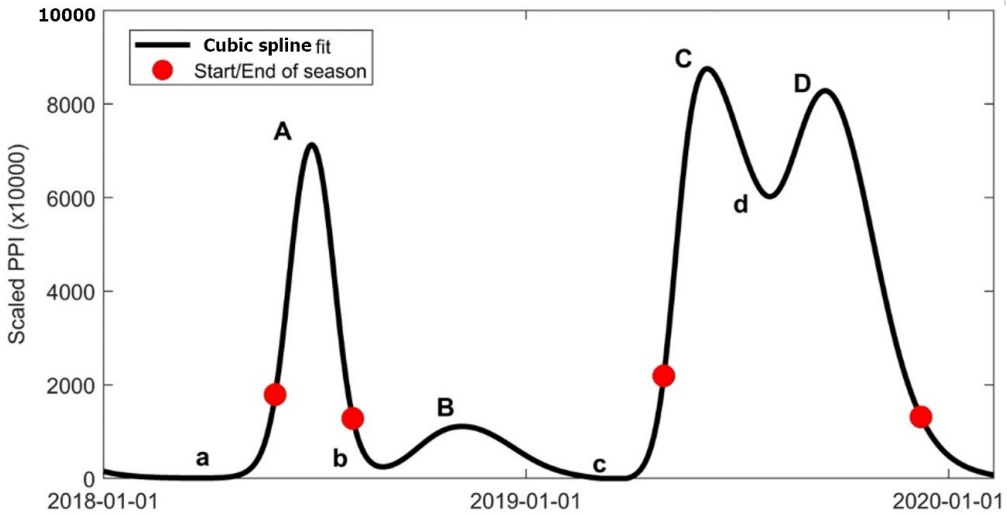


Figure 8. Example of the determination of seasons. The points, A, B, C, and D, are the peaks of coarse seasons, and the points, a, b, c, and d, are the corresponding local minima of coarse season. Red circles represent the start and end of the seasons.

3.2.7.3.4 Defining growing season thresholds

TIMESAT determines the timing of start of season (SOSD) and end of season (EOSD) by applying a threshold value in relation to the seasonal amplitude. Once start and end dates have been defined, the other VPP parameters are extracted. In order to harmonize the start and end of season dates of CGLOPS LSP with those of HR-VPP and MR-VPP, an analysis was made across 500 sites in Europe (Figure 13 in Appendix 5.2) by alternating the amplitude values between 0 and 100%, generating SOSD and EOSD values for these threshold values and computing the root mean square difference (RMSD) between the derived values and those from HR-VPP and MR-VPP. As seen in Figure 9, there are significant differences between phenological parameters estimated from PROBA-V versus those of HR-VPP and MR-VPP. The amplitude threshold value that minimizes the RMSD is around 20% for both SOSD and EOSD. Considering that such a comparison is from non-BRDF corrected data and with uncertainties in both LSP products, this comparison is only rough. To be more consistent with the thresholds setting in both MR-VPP and HR-VPP products, we selected 25% for SOSD and 15% for EOSD.

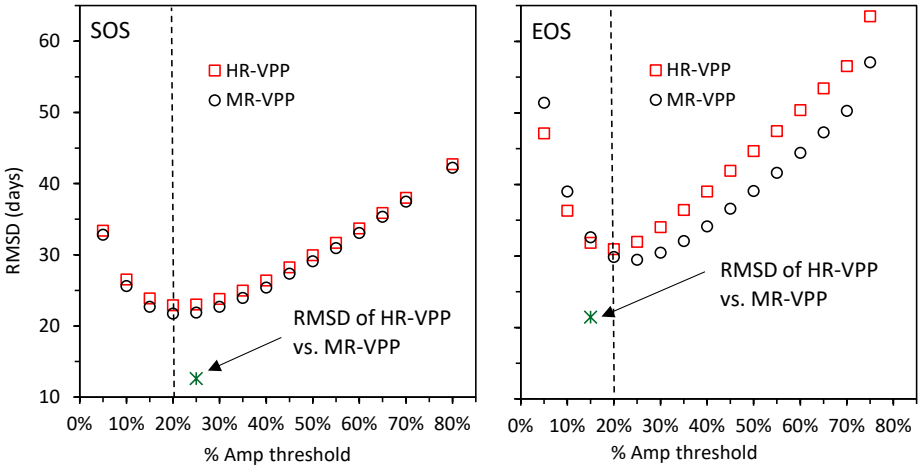


Figure 9. Analysis of RMSD between SOSD (left) and EOSD (right) for PROBA-V against HR-VPP and MR-VPP. These data are based on non-BRDF corrected daily TOC reflectance data over the common years of 2017, 2018, and 2019 for the three datasets across 500 EU sites.

3.2.7.3.5 Assigning seasons to years

Seasons may traverse years, thus a method for assigning them to the correct year is necessary. Seasons are assigned to years based on the season peak timing, e.g., season a-A-b is assigned to year 2018 and season c-C-d-D-e is assigned to year 2019 in Figure 8. In CGLOPS LSP, only two seasons are stored in a year. Therefore, annually only two seasons with the largest seasonal peaks are provided to users. The two seasons in a year are present chronologically, irrespective of their relative size. Thus, the main season may occur in Season 2 and a minor season in Season 1, or vice versa.

3.2.7.3.6 Defining thresholds for omitting seasons and regions

In three scenarios, the LSP are set to nodata-values (with QA=255) and the TIMESAT processing is skipped (BOX 1):

1. If the number of valid input data points throughout the entire time series is less than 3, or 1% of the total number of the data points. This primarily applies to areas lacking valid observations.
2. If the first-order PPI difference of the time series is too small, with fewer than 5 points exhibiting significant differences (>1e10^{-6}). This indicates weak seasonality, making it difficult to precisely determine phenological parameters, and mainly occurs with sustained evergreen vegetation.
3. If the number of points with PPI values above 2% of the peak value is less than 5 points. This primarily filters out inland water bodies with low PPI values occasionally exhibiting few extreme values possibly caused by noise.

BOX 1 Pseudo script for omitting seasons and regions

```
! Default.
process = True

Scenario 1
! total_npt:  total number of points
! y: PPI time series of 5-day interval
if count[points with weight > 0] < max(3, total_npt/100), process =
False

Scenario 2
if count[diff(y) > 1.d-6] < 5, process = False

Scenario 3
if count[y > (0.02*peak_value)] < 5, process = False
```

3.2.7.3.7 VPP QA

VPP Quality Assurance (QA) is assigned based on both the average of PPI-BRDF-weights (defined from the PPI DFlag in Table 3) and the total number of valid observations in each of phenological phases: green-up, green peak, and green-down of the growing season. They are defined as the left 20% – left 80%, left 80% – right 80%, and right 80% – right 20% of the season amplitude respectively (Figure 10). The overall QA is determined based on the leverage of the quality of the three phases.

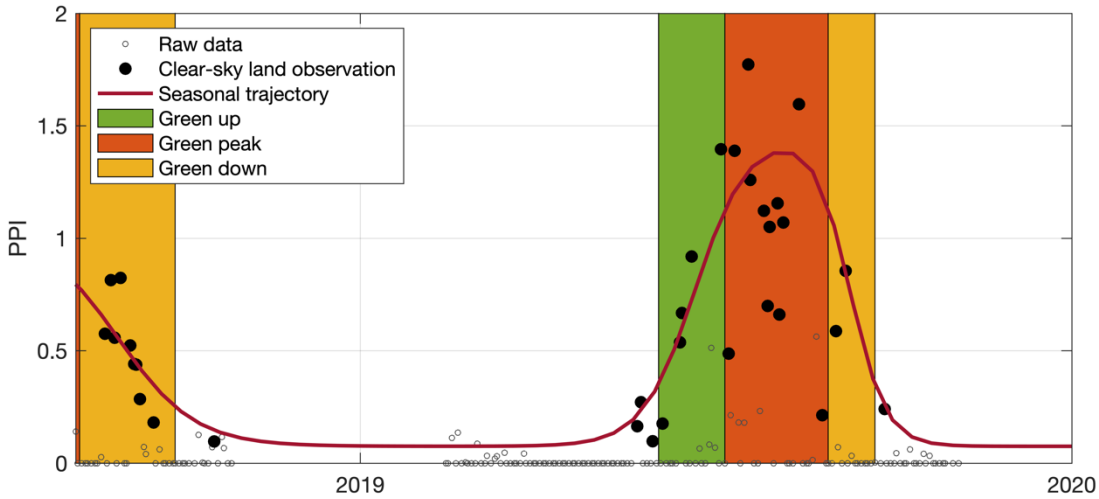


Figure 10. Example of VPP parameters QA.

Eight-bit (1 byte) QA values are generated as specified in Figure 11, and the details of bit values and descriptions are listed in Table 4. The QA flag is written to one output file for each season.

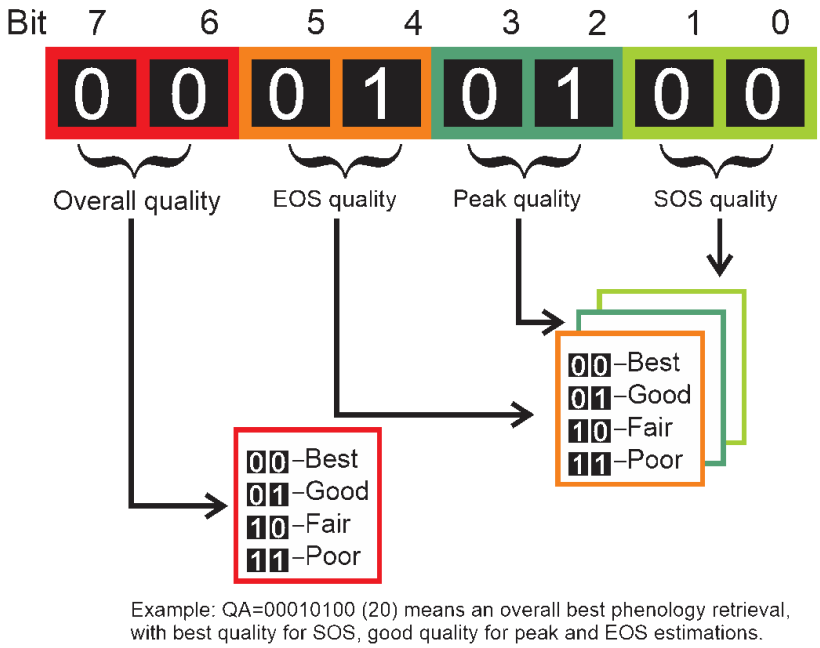


Figure 11. Details of 8-bit quality flags for phenology retrieval over a season.

Table 4. LSP QA flags and descriptions of phenology quality.

QA byte	Quality	Description
		SOS/Peak/EOS quality (two-bit values)
0	Best	More than 2 valid observations and average best PPI quality
1	Good	More than 2 valid observations and average good PPI quality
2	Fair	More than 2 valid observations and average fair PPI quality
3	Poor	Average poor PPI data quality, or fewer than 3 observations, or more than > 24 observations*
		Overall quality (two-bit values)
0	Best	No poor quality in three phenology phases
1	Good	Maximum one phase with poor quality
2	Fair	Maximum two phases with poor quality
3	Poor	All three phases with poor quality**
		Summary of QA (byte values)
QA <127	Good	Overall good quality, no more than one poor results in three phases
QA >127	Poor	Overall poor quality, more than one poor results in three phases
QA=255	Failure	No input, or no phenology estimation, or all poor estimates

Notes:

* Includes three situations: 1) fewer than 3 valid observations, 2) phenology phase shorter than 10 days, and 3) phenology phase longer than 4 months.

** Overlaps AQ=255 for situations of no inputs, or no phenology estimation. In any case, the LSP retrieval is considered a failure.

3.3 ASSESSMENT OF THE ALGORITHM PERFORMANCE

A summary of the quality assessment against various LSP products from other satellite platforms during the development phase is provided here. Further details are available in Appendix 5.3. The results of the exhaustive validation of the LSP product are available in [CGLOPS1_QAR_LSP-V1].

3.3.1 Cross-comparison of CGLOPS LSP against LSP from other satellite platforms

Appendix 5.3.1 provides information about relationships between CGLOPS LSP and LSP from other satellite platforms. A summary is given here.

The analysis was primarily aimed at comparing CGLOPS LSP against two high resolution LSP products: HR-VPP over Europe, and MuSLI LSP over North America. 500 test sites were selected over each of the two continents (Figure 13 and Figure 16). Further comparisons were made with medium resolution satellite LSP products, including MODIS MCD12Q2, VIIRS VNP22Q2 global phenology products, and EEA MR-VPP product. We also compared CGLOPS Sentinel-3 LSP 2019 against data from test sites for Europe and North America.

The comparison revealed strong correlations of PROBA-V and Sentinel-3 LSP with high-resolution regional LSP products, indicating consistent retrieval by the CGLOPS LSP processing chain and the high quality of input NBAR data. Specifically, PROBA-V demonstrated high correlations with HR-VPP for SOS (r=0.76) and EOS (r=0.88, Figure 14), and with MuSLI for SOS (r=0.77) and EOS (r=0.65, Figure 17). Additionally, it showed strong correlation with MR-VPP and MCD12Q2 for SOS (r=0.72~0.81) and EOS (r=0.67~0.92, Figure 14 and Figure 17). Similarly, Sentinel-3 exhibited even better correlations with other datasets, showing strong agreement with HR-VPP for SOS (r=0.83) and EOS (r=0.89, Figure 14), and with MuSLI for SOS (r=0.82) and EOS (r=0.78, Figure 17). Additionally, it displayed strong correlations with MR-VPP and MCD12Q2 for SOS (r=0.76~0.89) and EOS (r=0.80~0.93, Figure 14 and Figure 17). The SOS correlations of PROBA-V and Sentinel-3 with VIIRS VNP22Q2 were r=0.78~0.88 over European and r=0.71~0.76 over North American test sites, and the EOS correlations were r=0.75~0.78 over European and r=0.54~0.61 over North American (Figure 14 and Figure 17).

Productivity comparisons particularly highlighted consistency among PPI-based LSP datasets (MR-VPP, HR-VPP, PROBA-V, Sentinel-3, Figure 15), with discrepancies attributed to processing algorithms (spline fitting (SP) in this product vs. double logistic fitting (DL) in previous products). The productivities indicated by PPI sums retrieved using the cubic spline fitting algorithm applied to PROBA-V and Sentinel-3 data exhibit the highest correlations (r=0.94 in Europe and r=0.97 in North America). The EVI2 sums from MuSLI, MCD12Q2, VNP22Q2 varied significantly, with MuSLI exhibiting lower values compared to MCD12Q2 and VNP22Q2 (Figure 18).

Overall, we conclude that PPI-derived land surface phenology products compare well with other products, albeit with some noticeable scatter. Biases are high in some cases depending on differences in methodology and scales. The analyses underscore the potential of PROBA-V and Sentinel-3 LSP for phenological retrieval using NBAR reflectance and the TIMESAT 4.2 processing chain, while acknowledging methodological differences impacting dataset comparability.

3.3.2 Cross-comparison of PPI against LAI

Appendix 5.3.2 provides details about relationships between PPI and CGLOPS leaf area index (LAI). PPI generated with the described algorithm was compared against LAI data for a selection of sites globally, using a series of references, with 100 sites in each of 6 continents: Europe, Africa, Asia, North America, South America, and Oceanic (Figure 19). The comparison of PPI from daily PROBA-V TOC reflectance data and LAI, together with NDVI and EVI, the other two widely used vegetation indicators for phenology studies, are exemplified in Appendix 5.3.2.2 (Figure 20). The relationship between PROBA-V derived LAI and NBAR derived PPI are shown in scatter plots (Figure 21) with Pearson correlation coefficients shown in Table 6 for 12 major land cover types of these 600 test sites. These data indicate an overall better correspondence of PPI with LAI than of the two other indices (NDVI and EVI).

3.3.3 Selection of TIMESAT fitting method

The selection of the cubic spline method in favour of the double logistic fitting method was based on the high density of BRDF-corrected input data values and the method's superior ability to model PPI variability during the growing season. Additionally, the method generated more realistic estimates of seasonal parameters. For an example see Appendix 5.4.

3.4 Limitations of the algorithm

While PPI is a physically based index responding to green leaf area index (LAI) variations, it is based on some simplified assumptions. The vegetation canopy is treated like a turbid medium. In forest, violation of this assumption may be significant. Another assumption is that all vegetation is modelled with spherical (uniform) leaf angular distribution (G function in Eq. 4 is 0.5), but this may be violated in cases of erectophile (e.g. cereals) or planophile (e.g. potatoes) crops. Also, the reflectance of

vegetation with infinite leaf layers is retrieved from the highest value of time series data (MDVI) of relatively short time series data (a little more than 1 year), and the soil reflectance is assumed not to change with time. PPI is scaled between DVI of soil and a maximum DVI. These values represent pure soil and an infinitely thick canopy layer for a specific site. Random variations and noise in these values do not affect the value of PPI much during the start and the end of the season, but may affect the PPI values greatly during the peak growing season. The logarithmic expression in PPI makes high values sensitive to noise. Therefore, the index has been limited to 5.0 in the global context, and it is further being smoothed using TIMESAT. Furthermore, PPI is sensitive to BRDF effects; therefore, the test with non-BRDF corrected TOC reflectances only provided limited support.

In crop rotation schemes, the assumption of fixed level of maximum DVI is violated, which could lead to underestimation of the seasonal trajectory for the sparser crop. It has been noted that PPI is less sensitive to sparse vegetation than e.g. NDVI. This may lead to underestimation of winter crops or cover crops before the onset of the main growth period in spring time.

In sparsely vegetated areas, reaching a MDVI value representing a sufficiently thick leaf layer is unattainable, making it impossible to estimate a correct MDVI value from the long period of DVI time series. Therefore, we opt to assign a MDVI value of 0.18 to pixels with MDVI values below this threshold. This value is derived from the most frequently occurring (mode) MDVI value observed in the global dataset (PROBA-V NBAR reflectance 2018~2019). However, it is worth noting that this default MDVI value may need adjustment to better align with specific environmental conditions associated with different land cover types. Such refinement would contribute to further enhancing the spatial consistency of PPI-based LSP products, representing another stride toward achieving cross-platform consistency, as demonstrated in Appendix 5.3.1.

Data gaps in winter are filled by the base level value, which may hide some winter crops of low PPI from being detected. Although the smoothing spline can adapt to situations of irregular seasons, there is still no guarantee that all local variation will be detected.

The gap detection method could influence the coarse season detection due to data availability. An extreme case is when there are only few good observations with high-quality PPI values in a window larger than 91 days, leading to seasons being determined by these few data points. To address such cases, we assigned a poor quality (QA="11") for the corresponding phenology metrics determined by fewer than 3 valid data points. Caution should be taken when using phenology metrics with poor QA.

Since the outputs only allow a maximum of two seasons per year, some minor seasons may be missed if there are more than two cycles per year, which means that the outputs may not present all cycles, e.g., in the case of multiple moving events in grasslands.

During the development phase, access to NBAR data was crucial for minimizing biases in LSP resulting from observation geometry. However, due to limited availability of NBAR data, internal

validation of the CGLOPS LSP generation procedure relied on four distinct datasets: 1) LSP from the combined LAI dataset (combining PROBA-V and Sentinel-3) spanning from 2014 to 2022; 2) LSP from PPI derived from PROBA-V TOC reflectance for the years from 2014 to 2020 without BRDF correction; 3) LSP from PPI using a short time-series of BRDF-corrected PROBA-V TOC reflectance covering July 2018 to February 2020, supplemented by selected target tiles with NBAR data from October 2013 to December 2020; and 4) LSP from a brief time-series of BRDF-corrected Sentinel-3 data spanning from December 2018 to January 2020, focused on Europe, and from July 2018 to February 2020, focused on North America. It is important to acknowledge that the use of short time series of NBAR data and a limited test area may potentially impact the quality of the final product, and the tests with LAI and non-NBAR data only provide limited support for the internal evaluation results.

3.5 RISK OF FAILURE AND MITIGATION MEASURES

The estimation of LSP critically depends on the availability and quality of input data. Loss of accuracy in any of these data will degrade the output products. In the event of a failure in Sentinel-3 input data, one potential mitigation strategy involves transitioning to data from VIIRS. However, this will require a new round of calibrations and validation of the algorithm performance.

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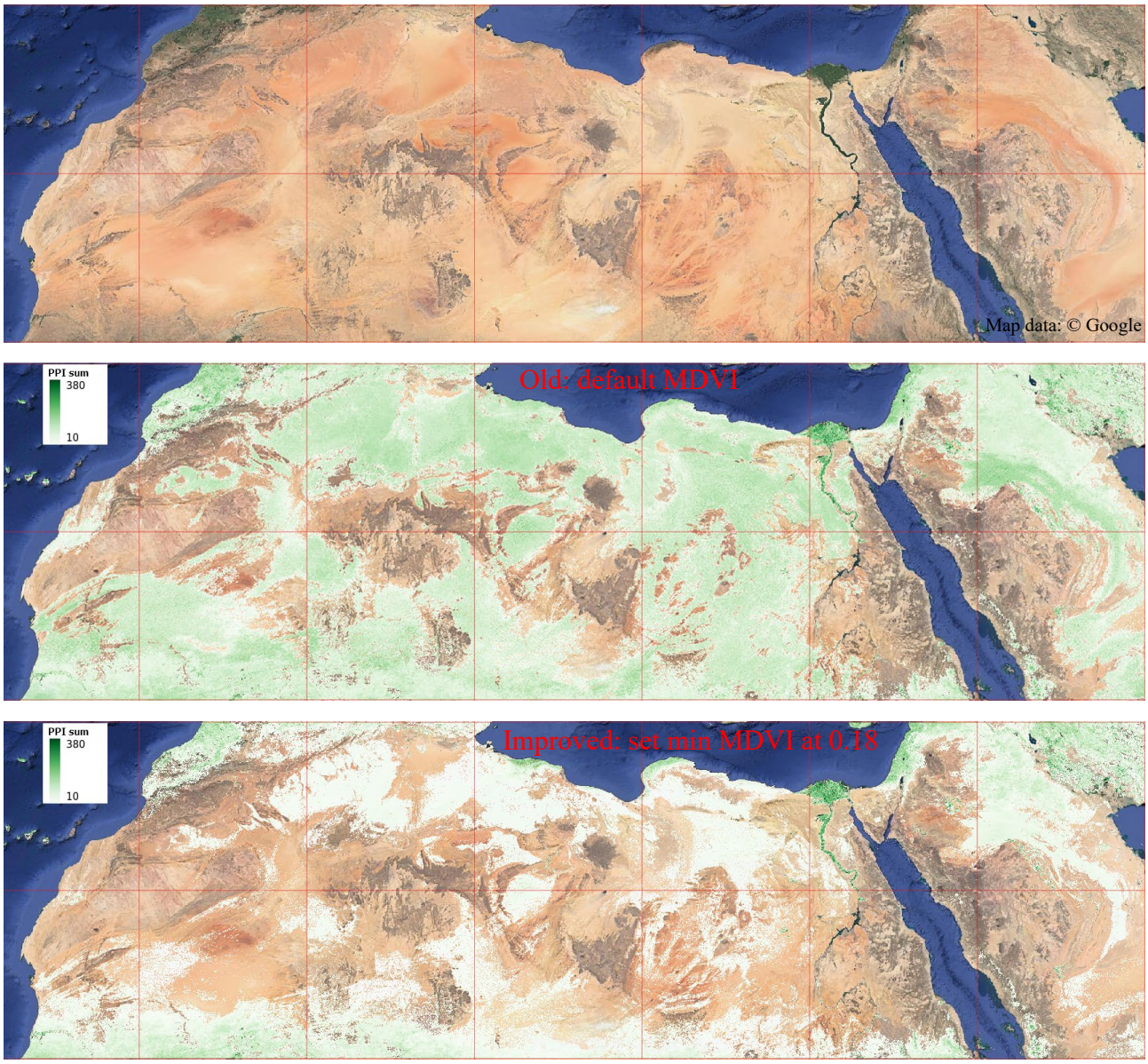
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5 APPENDIX

5.1 COMPARISON OF MDVI SETTING FOR SPARSE REGIONS

Figure 12 shows maps of the PPI sum in 2019 over the Sahelian and Arabian Deserts derived from two MDVI settings. The PPI sum appears unrealistically inflated when using the MDVI from the maximum DVI for individual pixel during January 2018 to February 2020. Conversely, when setting MDVI as 0.18, which represents the globe mode MDVI value, the PPI sum of 2019 presents a map comparable with land covers identified using Google Maps.



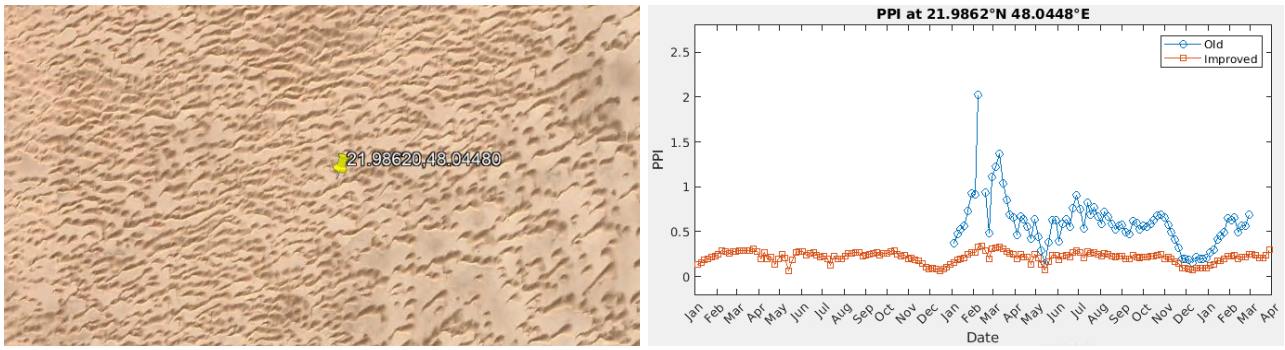


Figure 12. Maps of the PROBA-V PPI sum during the 2019 growing season, with MDVI set to 0.18, compared to default MDVI values. Top: Google Maps; Middle: default settings; Bottom: improved settings. The bottom graph shows an example of time-series data extracted over the Sahelian and Arabian Deserts.

5.2 SPECIFICATION OF OUTPUT PPI DATA

Table 5 shows specifications of the input PPI data used for phenology parameter estimation. Note that these data are for internal evaluation, not delivered to users.

Table 5. PPI data format and data quality flag (DFlag).

Name	Data type	Scale, offset	Data range	Fill value	Description	Unit
PPI	INT16	0.001,0	1000~5000	32767	PPI data	$m^2 \cdot m^{-2}$
DFlag	UINT8	-	1~252	255 for no observation 254 for MDVI<0.09 or isinf(PPI) 253 for unsuccessful ReBeLS inversion in NIR (QUIL=2)	Quality flag	-

5.3 COMPARISON WITH MULTIPLE PHENOLOGY PRODUCTS

In this analysis, the aim was to (1) explore PROBA-V-derived phenology using plant phenology index (PPI) and TIMESAT software, to be parameterized and validated using the **CLMS HR-VPP** product from Sentinel-2; (2) compare PROBA-V phenology with other phenology products, including **EEA's MODIS MR-VPP**, **NASA's MCD12Q2**, **VIIRS VNP22Q2**, and **Multi-Source Land Imaging (MuSLI) Land Surface Phenology (LSP)**; and (3) compare PPI with the **PROBA-V LAI** product. The first and second aims were carried on 500 selected sites in Europe and 500 selected sites in North America, the third on six continents: Europe, Africa, North America, South America, Asia, and Oceania, with 100 selected sites in each continent. This test investigated PPI and phenology using both PROBA-V daily TOC reflectances, and 5-day NBAR data. The data compared are pixel averages around the test sites with 3 × 3 pixels for PROBA-V and Sentinel-3 (nominal 1 km × 1 km), 91 × 91

pixels for HR-VPP and MuSLI and 2 × 2 pixels for MODIS and VIIRS derived datasets (MR-VPP, MCD12Q2 and VNP22Q2).

5.3.1 Comparing land surface phenology datasets over Europe and North America

5.3.1.1 500 LSP calibration sites for Europe

The data set contains 500 European calibration sites extracted to match the location of data sites in several programs: BELMANIP2.1 validation sites in the OLIVE program and Fluxnet2015 sites (Figure 13). Sources:

- BELMANIP 2.1 sites: https://calvalportal.ceos.org/web/olive/site-description
- Flux2015 sites: https://fluxnet.org/data/fluxnet2015-dataset/

Each test site is chosen in an area with relatively homogenous land cover. The homogenous areas were defined as a point from the above validation sites with homogenous landcover within a range of 5 x 5 pixels (nominal 1.5 x 1.5 km) identified from ESA CCI land cover in 2020.

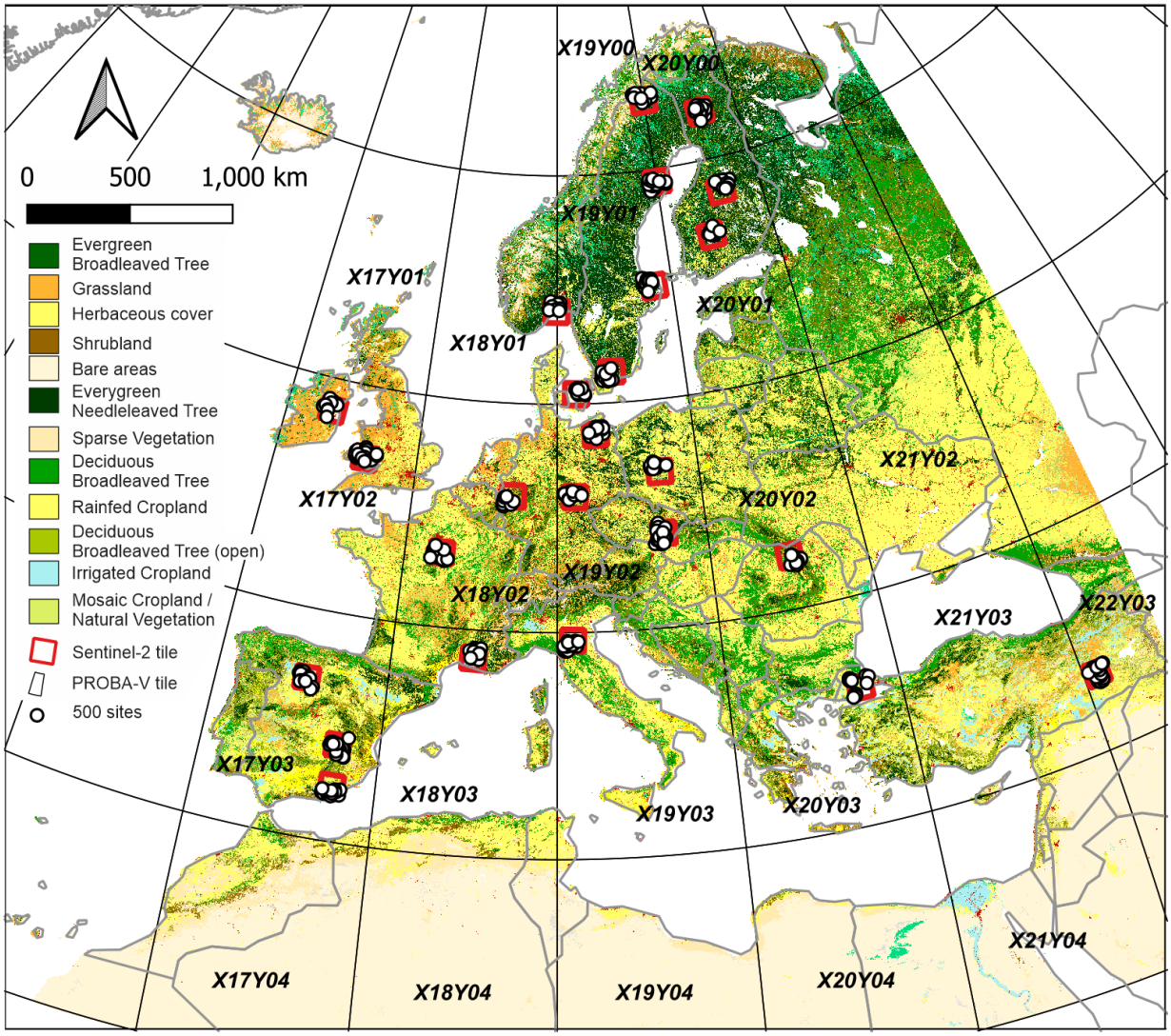


Figure 13. LSP calibration sites over Europe, n=500, overlaid on the ESA CCI Land cover 2020.

5.3.1.2 Comparison of LSP over 500 European calibration sites

Figure 14 shows the inter-comparison of the start-of-season (SOS) and end-of-season (EOS) dates from six phenology datasets: Sentinel-2 HR-VPP, MODIS MR-VPP, PROBA-V LSP, Sentinel-3 LSP, MODIS MCD12Q2, and VIIRS VNP22Q2. Among all the comparison pairs, the highest correlation is found between Sentinel-3 and PROBA-V for SOS ($r = 0.89$, $p < 0.001$) and EOS ($r = 0.94$, $p < 0.001$), with the lowest root-mean-square deviation (RMSD, SOS 13.6 days, EOS 15.7 days). These results support the feasibility of extending the PROBA-V processing chain to Sentinel-

3 data. High correlations (r = 0.74~0.93, p<0.001) also exist in comparisons among PROBA-V (Sentinel-3), MR-VPP and HR-VPP, which are all derived from the PPI vegetation index. The SOS and EOS from PROBA-V (Sentinel-3) have also high correlation with those from MODIS MCD12Q2 and VIIRS VNP22Q2 (r = 0.74~0.88, p<0.001). Both MCD12Q2 and VNP22Q2 used EVI2 to derive LSP, and their correlation over the 500 European test sites reaches 0.85 for SOS, and r=0.83 for EOS.

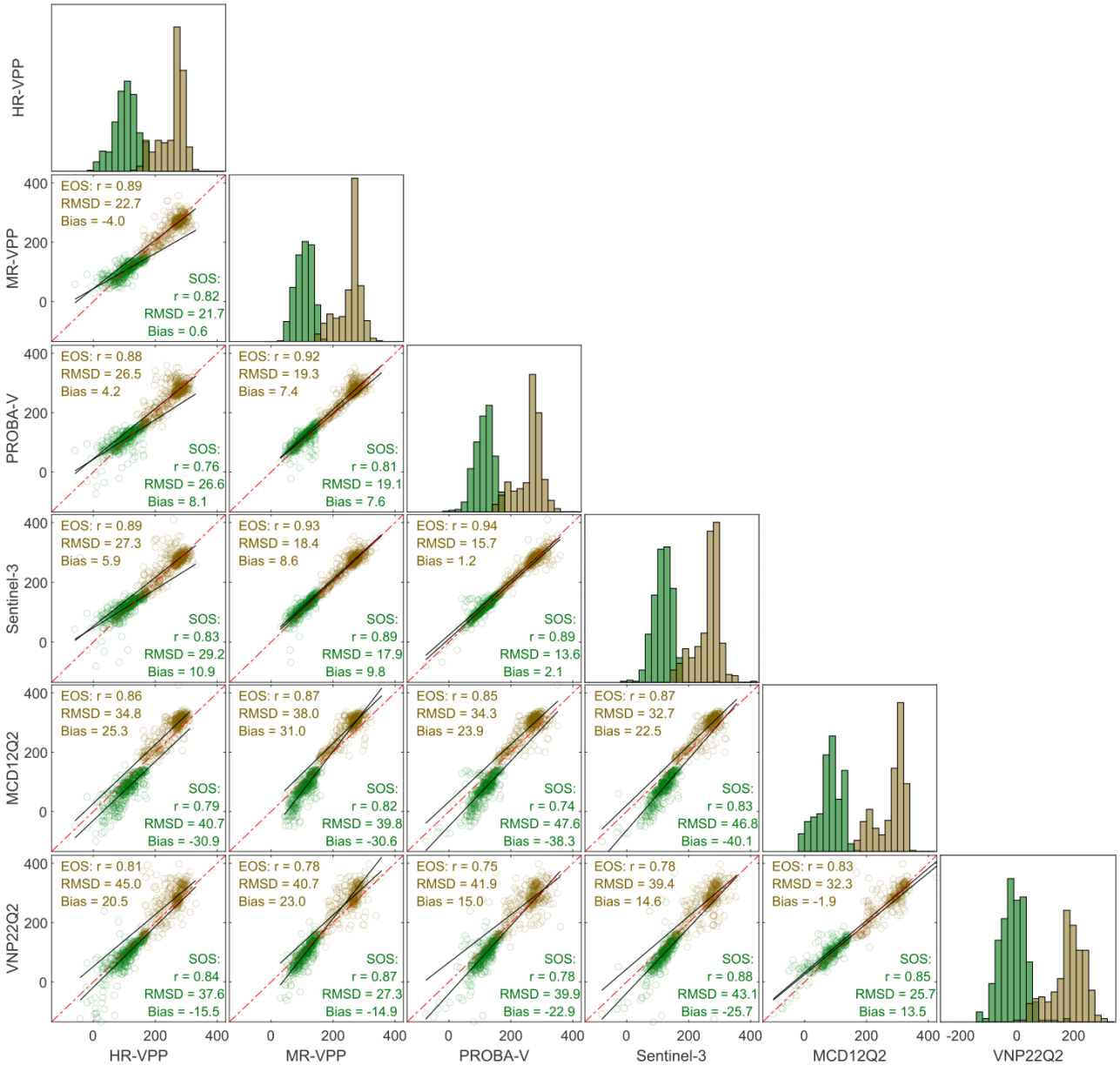


Figure 14. Comparison of start-of-season (SOS) and end-of-season (EOS) dates in 2019 over 500 European test sites from different LSP products. Unit: day-of-year (DOY).

Figure 15 shows the comparison of productivity (VI sums) during the growing season among all six LSP datasets. The VI sums show high consistency among the four PPI-based LSP datasets (MR-VPP,

HR-VPP, PROBA-V and Sentinel-3), with r = 0.83~0.94. Note that PPI sums shown in MR-VPP and HR-VPP are lower than those from PROBA-V and Sentinel-3 (scatter points above the 1:1 line in Figure 15). This is due to the improvement of the TIMESAT fitting using the cubic spline algorithm, having high fidelity to raw data and thus more accurate fit. The MR-VPP and HR-VPP adopted double logistic processing, which negatively biased the peak season productivity, leading to PPI sums that are likely lower than reality. The EVI2-based two NASA's LSP products, MCD12Q2 and VNP22Q2, produced largely deviating growing season EVI2 sums. VNP22Q2 estimated productivity 91% higher than that of MCD12Q2 (the average EVI2 sum is 33.0 in VNP22Q2, and 17.3 in MCD12Q2).

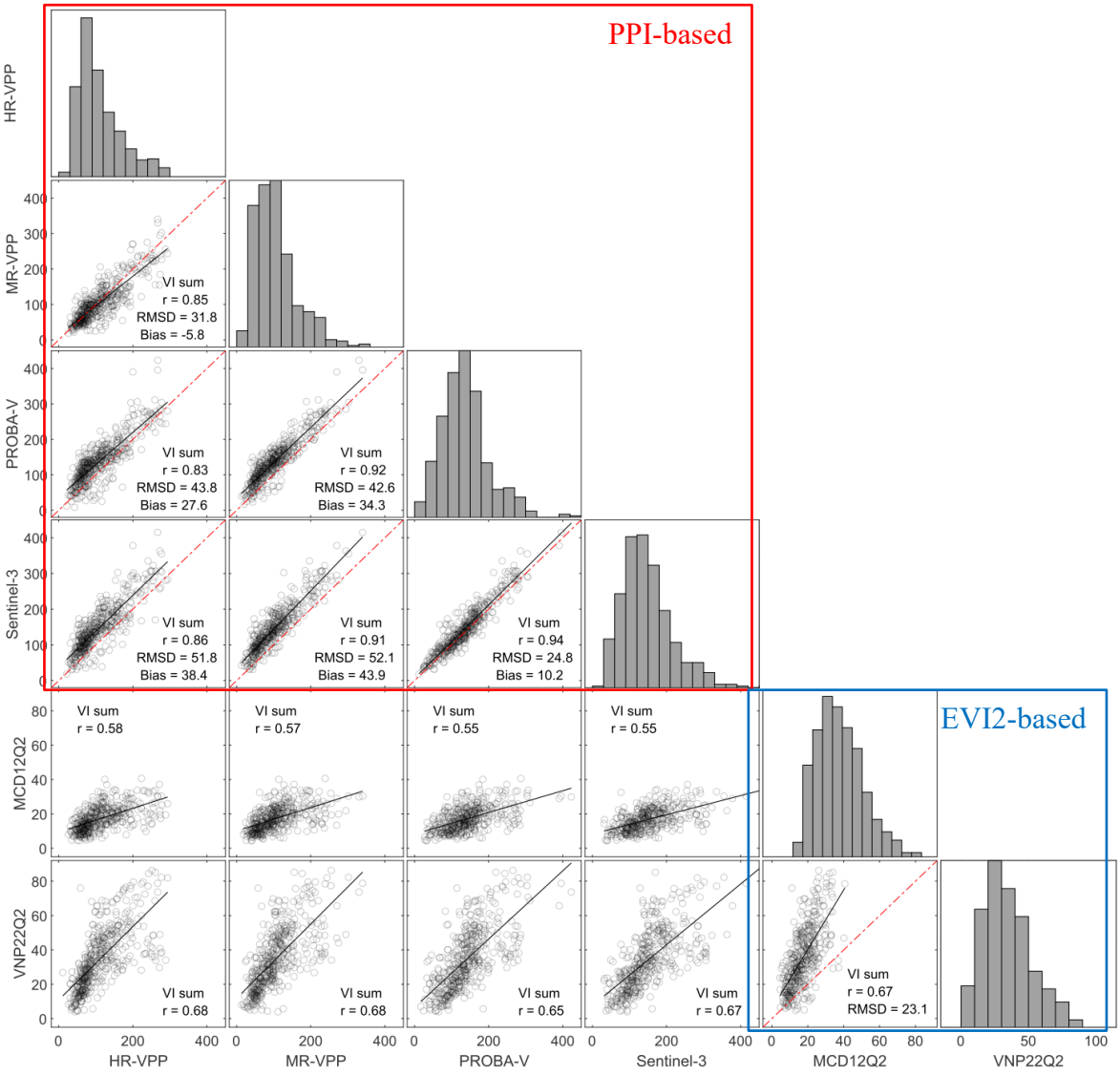


Figure 15. Comparison of growing season sum of VIs from different LSP products.

5.3.1.3 500 calibration sites for North America

This dataset contains 500 North American calibration sites overlapping BELMANIP2.1 validation sites in the OLIVE program and Fluxnet2015 sites (Figure 16). Sources:

- BELMANIP 2.1 sites https://calvalportal.ceos.org/web/olive/site-description
- Flux2015 sites https://fluxnet.org/data/fluxnet2015-dataset/

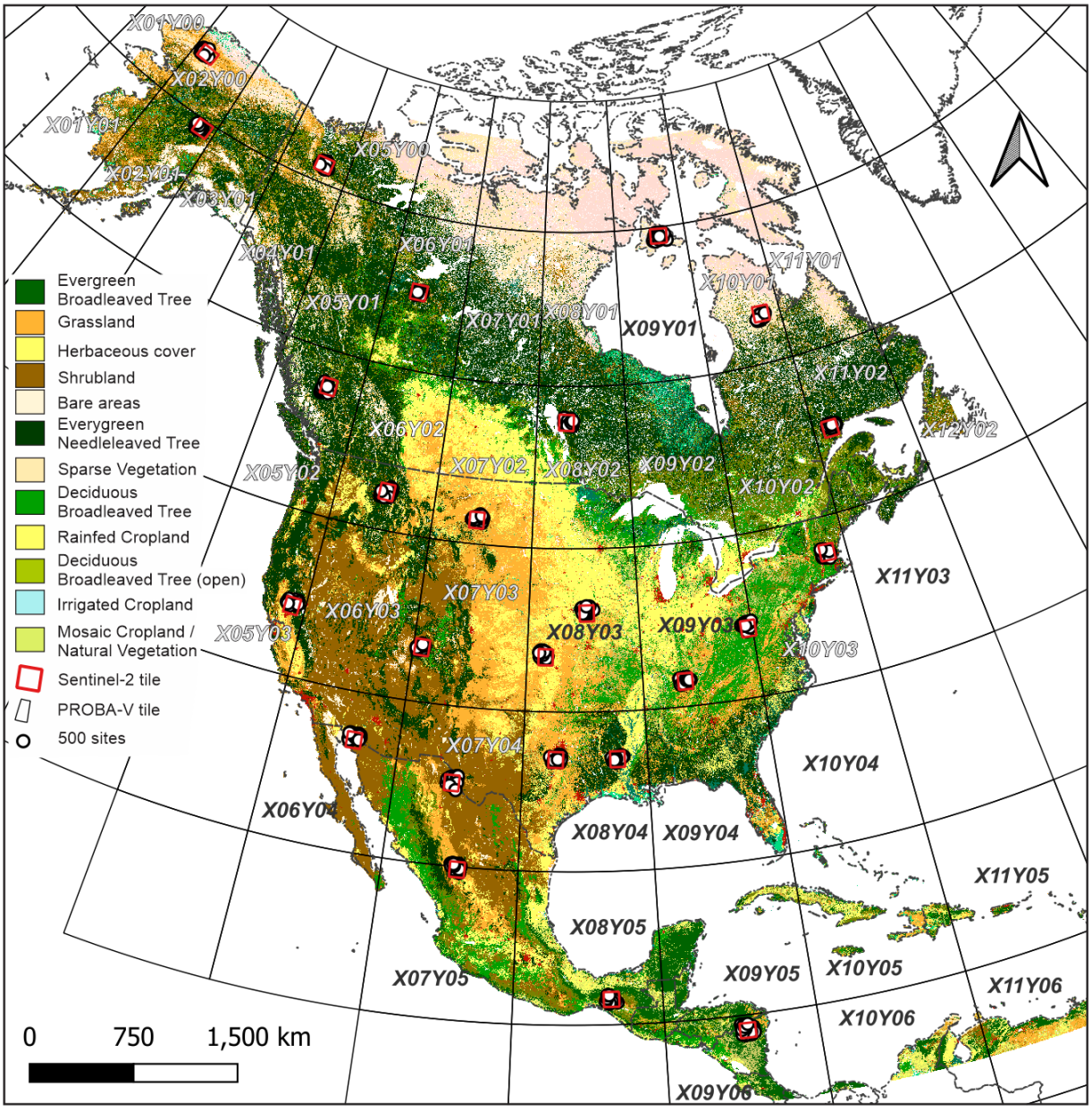


Figure 16. Calibration sites over North America, n = 500 overlaid on the ESA CCI Land cover 2020.

Each North American calibration site is located within homogeneous land cover, determined in the same way as for the European calibration sites.

5.3.1.4 Comparison of LSP over 500 North American sites

Figure 17 shows comparisons of SOS and EOS among five LSP datasets: PROBA-V, Sentinel-3, MuSLI, MCD12Q2, and VNP22Q2. The SOS dates derived from Sentinel-3 exhibit high Pearson correlations with those from other four platforms (r=0.76~0.82, p<0.001). The EOS dates from Sentinel-3 also have relatively high correlations with those of PROBA-V, MCD12Q2, and MuSLI (r=0.78~0.81, p<0.001) and relatively low correlation with VNP22Q2 (r=0.61, p<0.001).

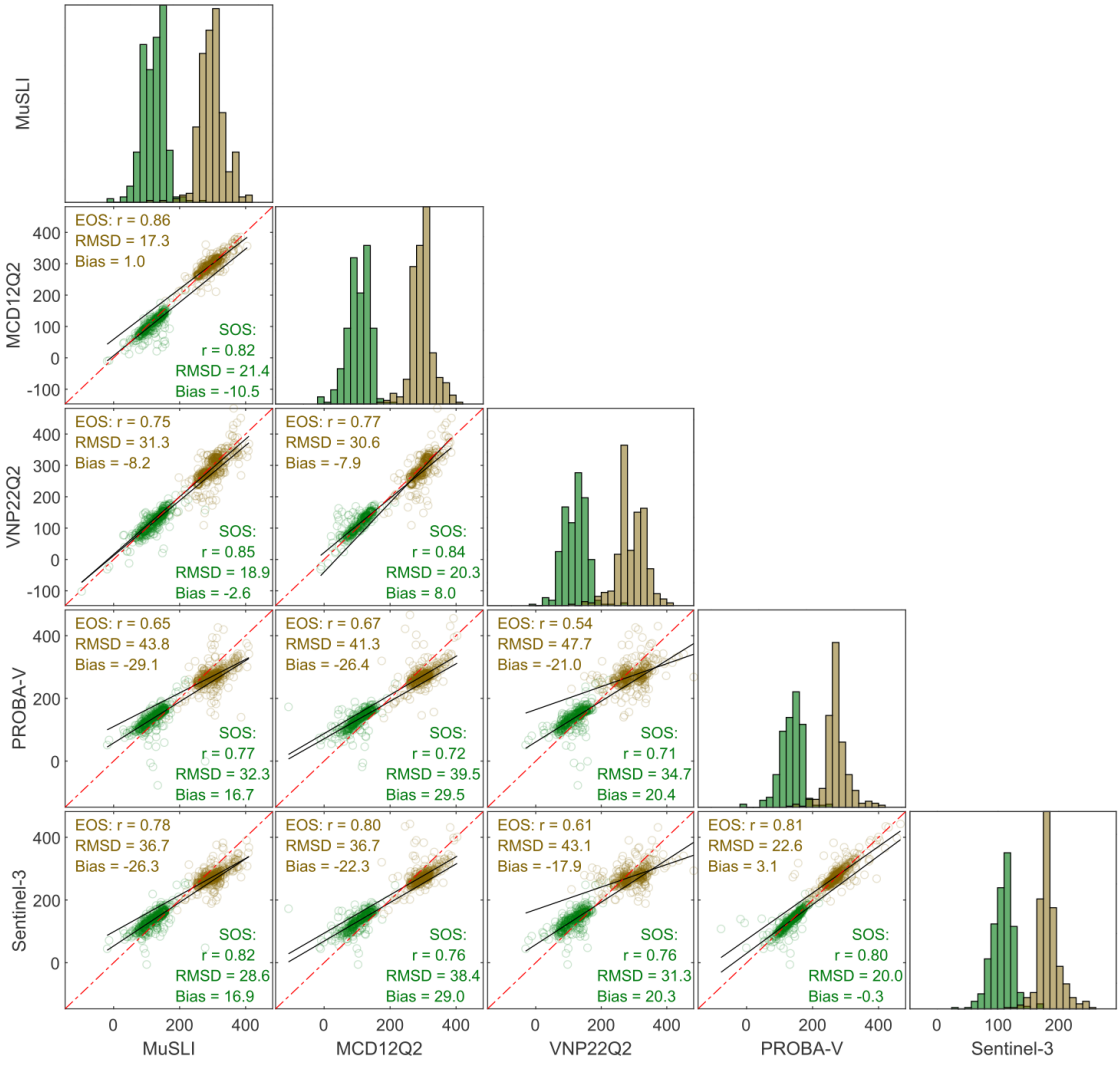


Figure 17. Comparison of start-of-season (SOS) and end of season (EOS) dates in 2019 over 500 North American test sites from different LSP products. Unit: date-of-year (DOY)

The phenology metrics from MuSLI and MCD12Q2 display the highest correlations among the four datasets in the comparison (SOS: r=0.82, p<0.001, EOS: r=0.86, p<0.001). Similarly, the phenology metrics from PROBA-V and Sentinel-3 also display high correlations with r=0.80, p<0.001 for SOS; and r=0.81, p<0.001 for EOS.

Figure 18 illustrates the comparison of VI sums among the four LSP datasets. Since PPI is expressed in units of m^2m^{-2} and EVI-2 has no unit, direct comparisons of the VI sums between PPI-based phenology and EVI2-based phenology are not feasible. We compare their correlations between PPI-based and EVI2-base productivities.

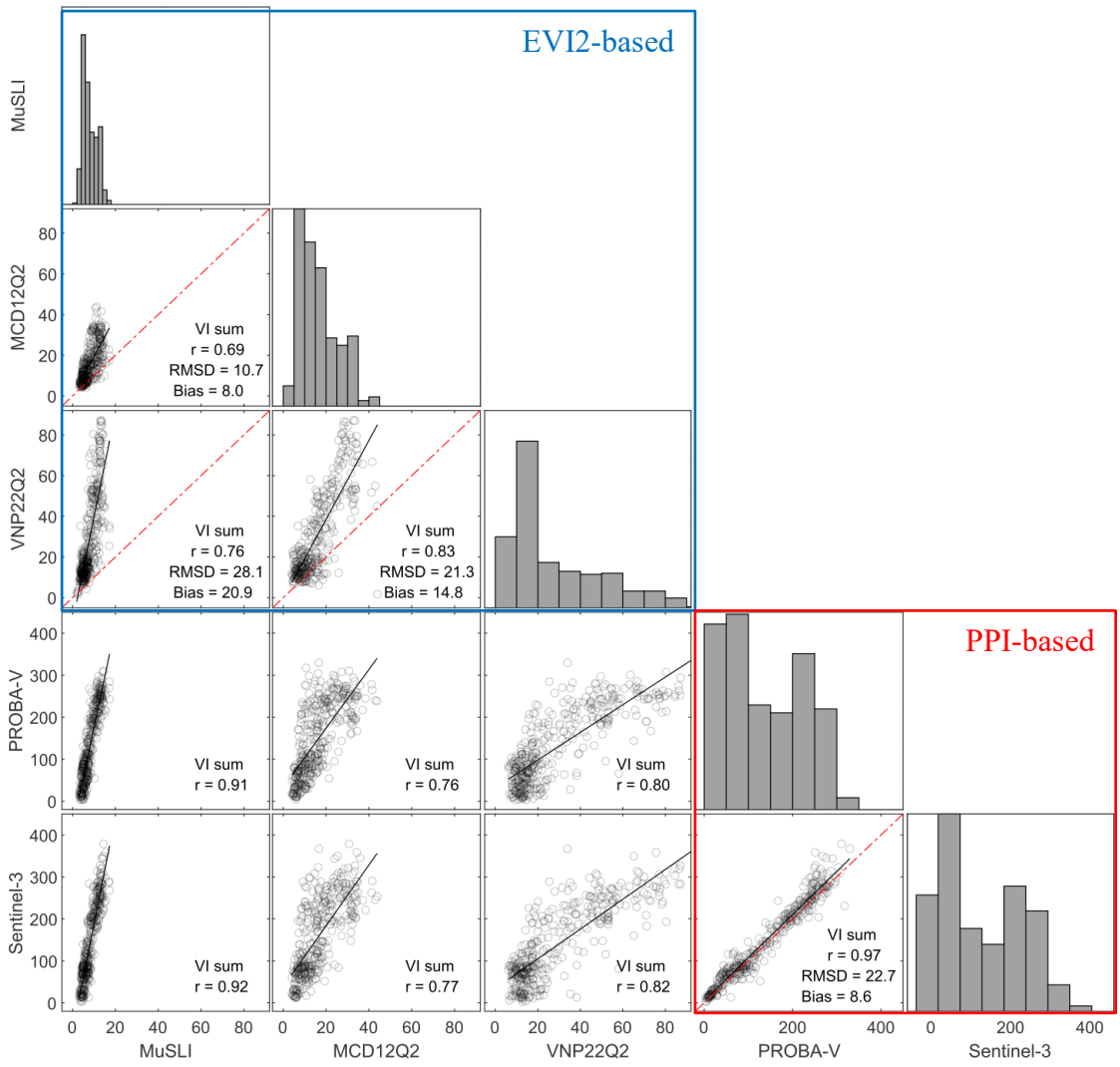


Figure 18. Comparison of seasonal productivity (growing season sum) of VIs from different LSP products.

The two PPI-based data sets PROBA-V and Sentinel-3 have the highest correlation in productivity (r=0.97, p<0.001). Moreover, the PPI sums from PROBA-V and Sentinel-3 demonstrate strong

Pearson correlations with the EVI2 sum from MuSLI LSP dataset (r=0.91~0.92, p<0.001). Additionally, the correlations of the EVI2 sums among the MuSLI, MCD12Q2, and VNP22Q2 datasets are also high (r=0.69~0.83, p<0.001). However, their magnitudes of EVI2 sum, even when derived from the same vegetation index, are not comparable. For instance, the EVI2 sum from MuSLI is on average 51% smaller than from MCD12Q2, and 72% smaller than that from VNP22Q2.

Altogether, we conclude that PPI derived land surface phenology products compare well with other products with some noticeable scatter. High biases and low correlations may exist in some cases, depending on differences in methodology and scales.

5.3.2 Comparison of LAI and PPI over global land cover classes

5.3.2.1 CGLOPS1 100 sites comparison over 6 continents

The 100 test sites on each of the 6 continents (North America, South America, Europe, Africa, Asia, and Oceania) were selected based on the following reference networks and datasets:

- BELMANIP2.1 (Global 445 sites) and DIRECT (Global 113 sites)
- LandVAL Network (Global 725 sites)
- Fluxnet 2015 (Global, 267 sites)
- ICOS flux sites in (Europe, 52 terrestrial ecosystem flux sites)
- NEON LAI sites (USA, 56 sites. Spectrometer mosaic, 2013-2021)
- Planet sites (USA, 100 sites, Planet satellite data)
- African Rangeland sites (African, 27 sites from the ESA RAMONA project)

The 100 continental calibration sites were selected not to overlap the other 500 sites in Europe and 500 sites in North America and are displayed in Figure 19.

Two evaluations were conducted to assess the relationship between PPI and LAI. First, we compared the three daily TOC reflectances derived VIs (PPI, NDVI and EVI) with LAI spanning from 2014 to 2019. This comparison involved testing the PPI computation using ideal solar diffuse fraction (standard atmospheric composition) and using ECMWF ERA5 reanalysed solar radiation product. The comparison encompassed 100 sites in Africa and Europe, in addition to the previous 500 sites in Europe. Second, after the PROBA-V NBAR 5-day reflectance was available over the entire globe for 2019, PPI was computed and compared with LAI for the year 2019. The comparison used scatter plots for different land cover types over all six continents.

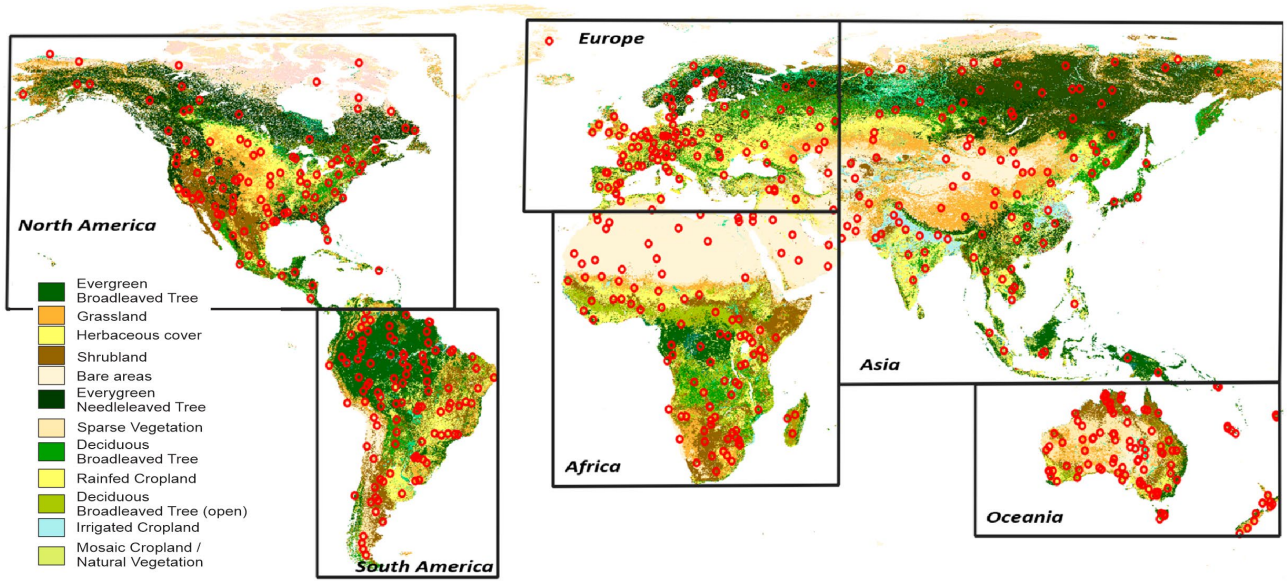
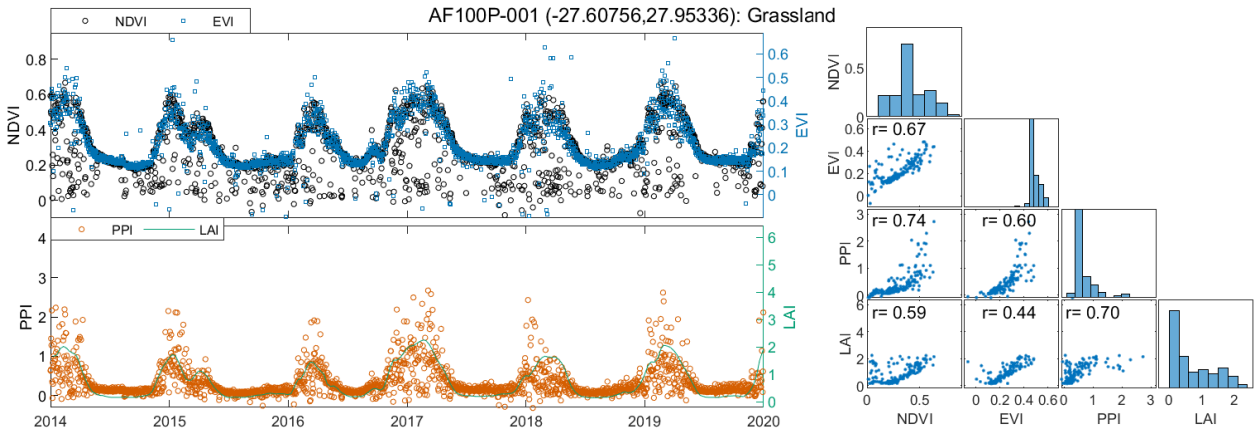
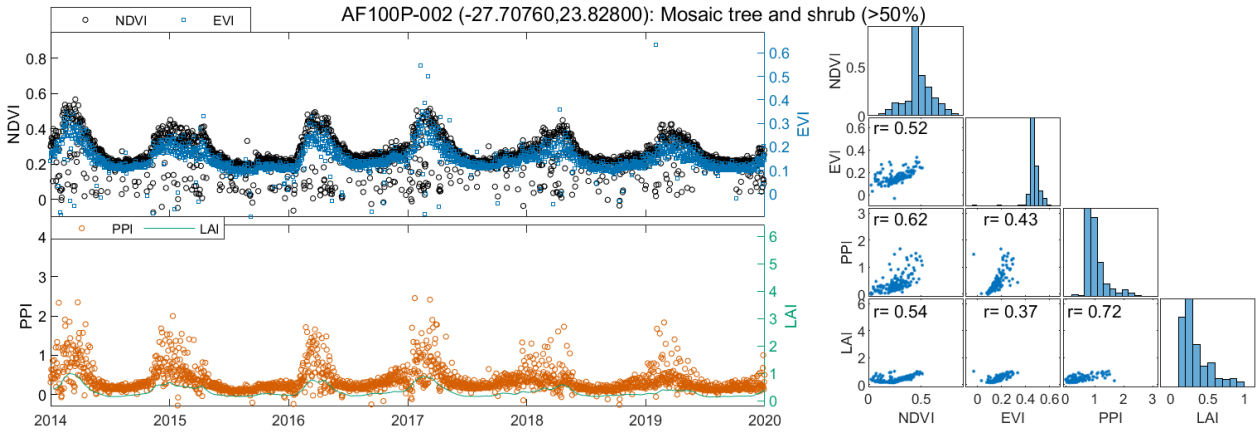


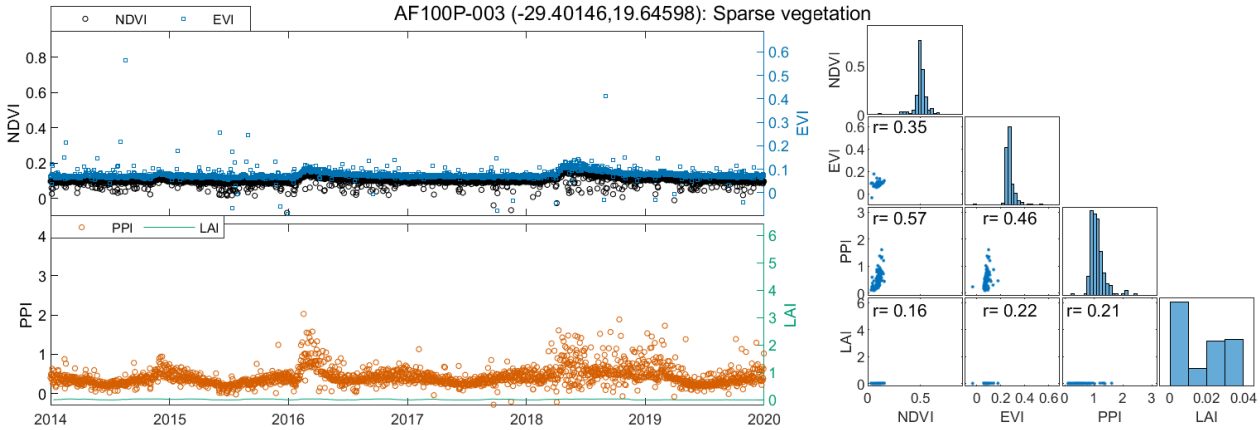
Figure 19. 100 calibration sites for Sentinel 3 LAI vs. PPI over 6 continents. The background land cover map is from ESA CCI LC 2020.

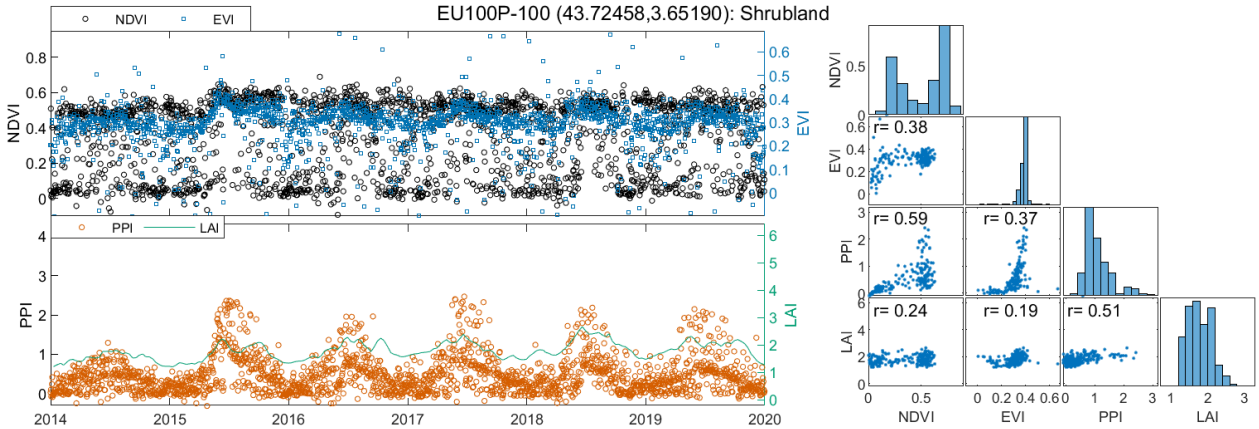
5.3.2.2 Comparison of daily PPI data (without NBAR correction) with LAI over African and European sites

Figure 20 presents several examples of the comparison results for PPI and LAI time series, along with NDVI and EVI time series. The scatterplots, histograms, and Pearson correlation coefficients among them are also shown in the figure. The time series data show that land covers with intuitively pronounced seasonality, such as grassland and deciduous forests, exhibit clear seasonal variations in all the three indices: PPI, EVI, and NDVI. Conversely, in evergreen forests and shrub lands, PPI demonstrates more noticeable seasonal fluctuations.









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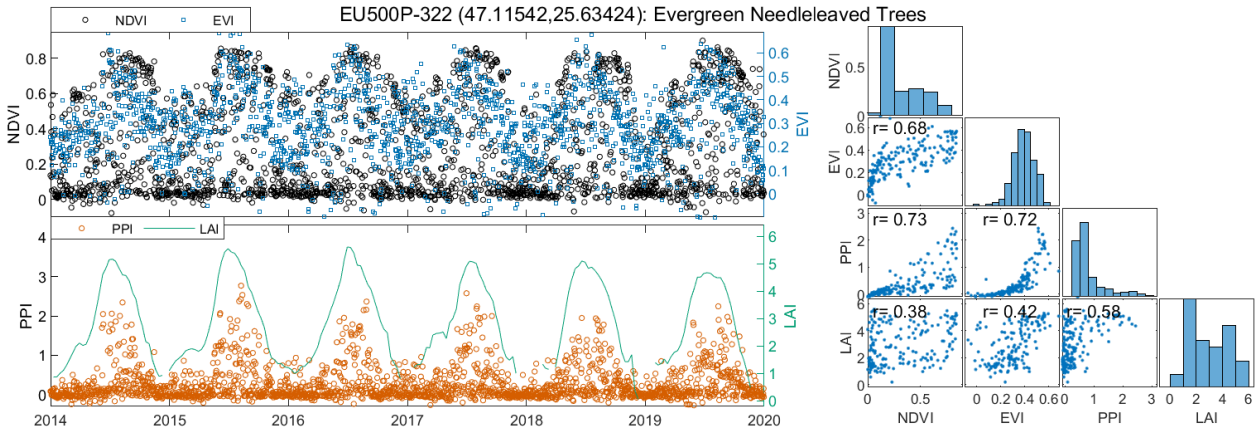


Figure 20. Examples of PPI, EVI, and NDVI time-series data for different land cover classes over Africa and Europe, and comparison with CGLOPS LAI. Pearson correlation coefficients (r) are shown in the scatter plots.

5.3.2.3 Comparison of 5-day PPI data (with NBAR correction) with LAI over 6 × 100 sites

Figure 21 shows the comparison between PPI and CGLOPS LAI across six continents, encompassing 100 sites each, representing a variety of 12 predominant land cover types. The proportion of each land cover type across the 600 test sites and the corresponding Pearson correlation coefficients are presented in Table 6.

The scatter plots illustrate a strong relationship between PPI and LAI in Grassland, Herbaceous cover, Cropland, and Deciduous Broadleaved Forest, albeit with some variability. However, the comparison between PPI and LAI is less favourable in Evergreen Broadleaved Trees and Deciduous Broadleaved Trees (close forest), particularly as LAI tends to saturate at approximately 6 m^2m^{-2}, after PPI exceeds 1 m^2m^{-2}. This is a documented limitation of PROBA-V LAI and has been corrected for Sentinel-3 LAI [CGLOPS1_ATBD_LAI300m-V1.1]. Further investigation into both PPI and LAI datasets is warranted when longer time-series of Sentinel-3 become available.

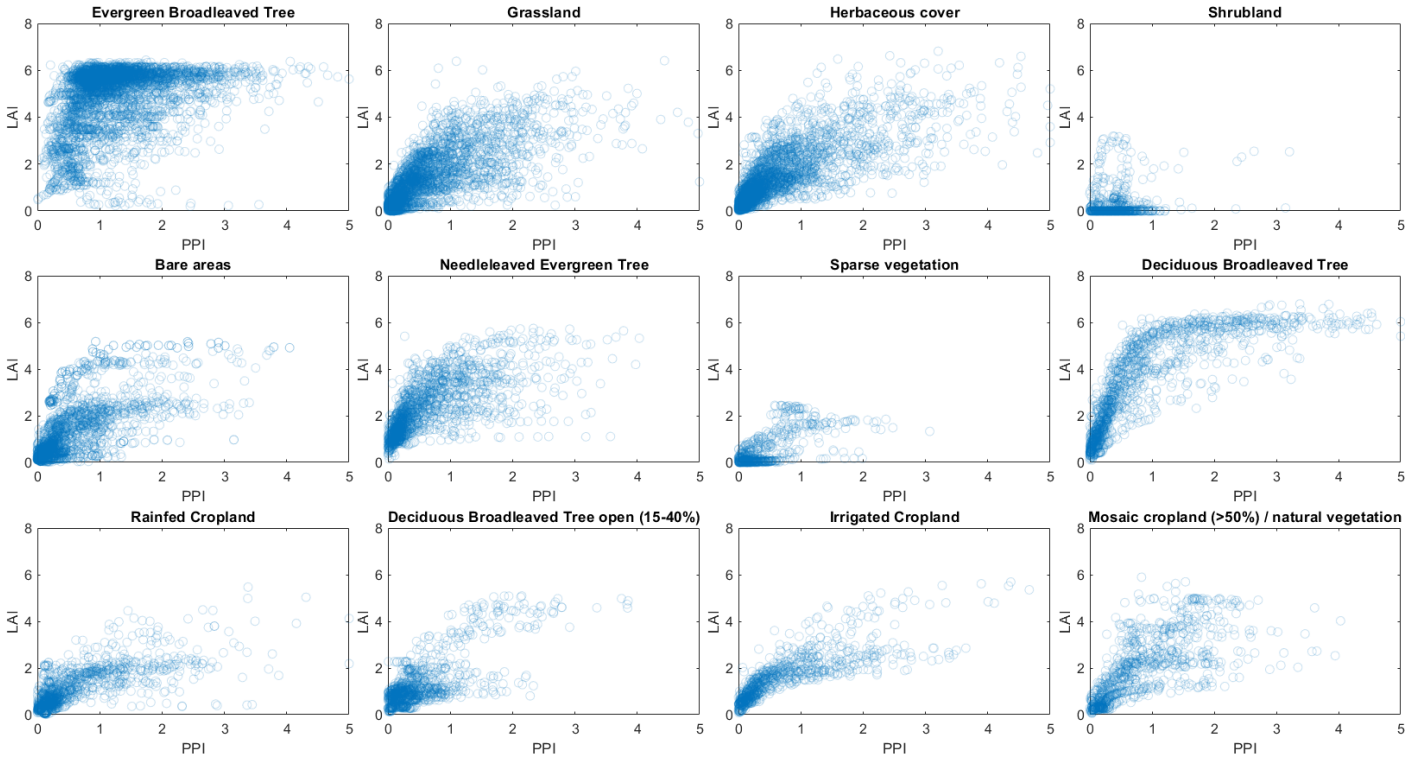


Figure 21. Scatter plots of PPI vs PROBA-V LAI in 2019 over 12 majority land cover types.

Table 6. The proportion of majority land cover type among 6×100 test sites and the correlation coefficients between LAI and PPI.

ESA CCI LC 2020	Name	Proportion	Correlation coefficient (p<0.001)
50	Evergreen Broadleaved Tree	12.7%	0.33
130	Grassland	12.3%	0.79
11	Herbaceous cover	10.5%	0.81
120	Shrubland	8.3%	0.30
200	Bare areas	8.0%	0.75
70	Evergreen Needleleaved Tree	6.5%	0.71
150	Sparse vegetation	6.0%	0.65
60	Deciduous Broadleaved Tree	5.3%	0.81
10	Rainfed Cropland	3.8%	0.74
62	Deciduous Broadleaved Tree open (15-40%)	3.0%	0.77
20	Irrigated Cropland	2.5%	0.83
30	Mosaic cropland (>50%) / natural vegetation (tree; shrub; herbaceous cover) (<50%)	2.5%	0.70

5.4 COMPARISON OF TIMESAT FITTING ALGORITHMS: DOUBLE LOGISTIC VS. CUBIC SPLINE

Figure 22 shows maps depicting the SOS, EOS, and PPI sum for the year 2018, as estimated using TIMESAT's double logistic (DL) and cubic spline (SP) fitting algorithms over southern Sweden (PROBA-V tile X19Y01). These estimations are compared with those of MR-VPP derived from MODIS.

In general, the SOS and EOS estimates obtained through the TIMESAT SP algorithm match well with those from MR-VPP. In contrast, the TIMESAT DL algorithm generates an SOS that is notably earlier and an EOS that is much later. The EOS generated by TIMESAT DL appears unrealistically delayed compared to that produced by the SP algorithm. For instance, in the case of cropland in southern Sweden in 2018, the EOS derived from the DL algorithm occurs at November 15, whereas it occurs at July 21 according to the SP algorithm.

Consequently, due to the more realistic timing of phenological events, the cubic spline fitting algorithm was selected for CGLOPS Land Surface Phenology product.

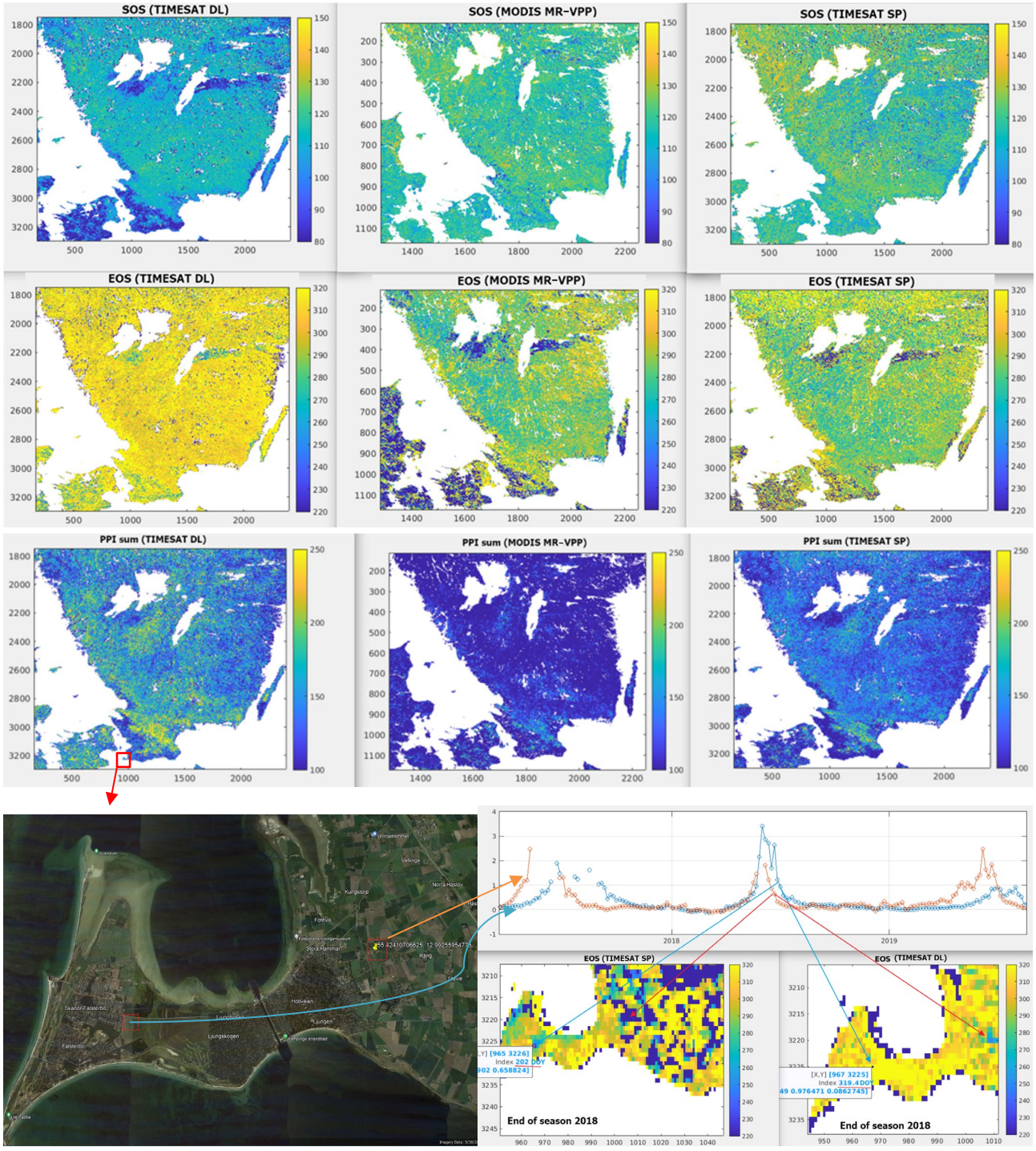


Figure 22. Examples of LSP retrieval in 2018 using TIMESAT Double Logistic (DL) algorithm vs Cubic spline algorithm from PROBA-V NBAR data, in comparison with MODIS MR-VPP. The relative thresholds SOS 25% and EOS 15% amplitude were adopted for all tests. Note that the projection of PROBA-V data is EPSG 4326 and for MODIS it is sinusoidal.